

Deep learning

In machine learning, **deep learning** (DL) focuses on utilizing multilayered neural networks to perform tasks such as classification, regression, and representation learning. The field takes inspiration from biological neuroscience and revolves around stacking artificial neurons into layers and "training" them to process data. The adjective "deep" refers to the use of multiple layers (ranging from three to several hundred or thousands) in the network. Methods used can be supervised, semi-supervised or unsupervised.^[2]

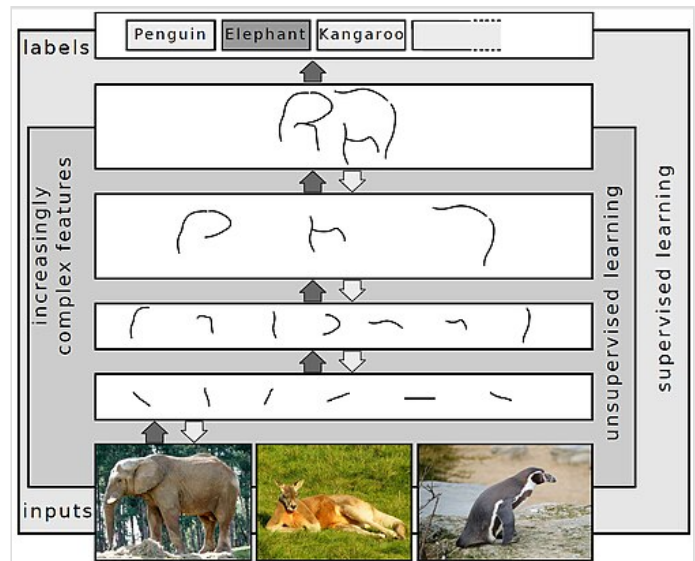
Some common deep learning network architectures include fully connected networks, deep belief networks, recurrent neural networks, convolutional neural networks, generative adversarial networks, transformers, and neural radiance fields. These architectures have been applied to fields including computer vision, speech recognition, natural language processing, machine translation, bioinformatics, drug design, medical image analysis, climate science, material inspection and board game programs, where they have produced results comparable to and in some cases surpassing human expert performance.^{[3][4][5]}

Early forms of neural networks were inspired by information processing and distributed communication nodes in biological systems, particularly the human brain. However, current neural networks do not intend to model the brain function of organisms, and are generally seen as low-quality models for that purpose.^[6]

Overview

Most modern deep learning models are based on multi-layered neural networks such as convolutional neural networks and transformers, although they can also include propositional formulas or latent variables organized layer-wise in deep generative models such as the nodes in deep belief networks and deep Boltzmann machines.^[7]

Fundamentally, deep learning refers to a class of machine learning algorithms in which a hierarchy of layers is used to transform input data into a progressively more abstract and composite representation. For example, in an image recognition model, the raw input may be an image (represented as a tensor of pixels). The first representational layer may attempt to identify basic shapes such as lines and circles, the second layer may compose and encode arrangements of edges, the third layer may encode a nose and eyes, and the fourth layer may recognize that the image contains a face.



Representing images on multiple layers of abstraction in deep learning^[1]

Importantly, a deep learning process can learn which features to optimally place at which level *on its own*. Prior to deep learning, machine learning techniques often involved hand-crafted feature engineering to transform the data into a more suitable representation for a classification algorithm to operate on. In the deep learning approach, features are not hand-crafted and the model discovers useful feature representations from the data automatically. This does not eliminate the need for hand-tuning; for example, varying numbers of layers and layer sizes can provide different degrees of abstraction.^{[8][2]}

The word "deep" in "deep learning" refers to the number of layers through which the data is transformed. More precisely, deep learning systems have a substantial *credit assignment path* (CAP) depth. The CAP is the chain of transformations from input to output. CAPs describe potentially causal connections between input and output. For a feedforward neural network, the depth of the CAPs is that of the network and is the number of hidden layers plus one (as the output layer is also parameterized). For recurrent neural networks, in which a signal may propagate through a layer more than once, the CAP depth is potentially unlimited.^[9] No universally agreed-upon threshold of depth divides shallow learning from deep learning, but most researchers agree that deep learning involves CAP depth higher than two. CAP of depth two has been shown to be a universal approximator in the sense that it can emulate any function.^[10] Beyond that, more layers do not add to the function approximator ability of the network. Deep models (CAP > two) are able to extract better features than shallow models and hence, extra layers help in learning the features effectively.

Deep learning architectures can be constructed with a greedy layer-by-layer method.^[11] Deep learning helps to disentangle these abstractions and pick out which features improve performance.^[8]

Deep learning algorithms can be applied to unsupervised learning tasks. This is an important benefit because unlabeled data is more abundant than labeled data. Examples of deep structures that can be trained in an unsupervised manner are deep belief networks.^{[8][12]}

The term *deep learning* was introduced to the machine learning community by Rina Dechter in 1986,^[13] and to artificial neural networks by Igor Aizenberg and colleagues in 2000, in the context of Boolean threshold neurons.^{[14][15]} The etymology of the term is more complicated.^[16]

Interpretations

Deep neural networks are generally interpreted in terms of the universal approximation theorem^{[17][18][19][20][21]} or probabilistic inference.^{[22][23][8][9][24]}

The classic universal approximation theorem concerns the capacity of feedforward neural networks with a single hidden layer of finite size to approximate continuous functions.^{[17][18][19][20]} In 1989, the first proof was published by George Cybenko for sigmoid activation functions^[17] and was generalised to feed-forward multi-layer architectures in 1991 by Kurt Hornik.^[18] Recent work also showed that universal approximation also holds for non-bounded activation functions such as Kunihiko Fukushima's rectified linear unit.^{[25][26]}

The universal approximation theorem for deep neural networks concerns the capacity of networks with bounded width but the depth is allowed to grow. Lu et al.^[21] proved that if the width of a deep neural network with ReLU activation is strictly larger than the input dimension, then the network can

approximate any Lebesgue integrable function; if the width is smaller or equal to the input dimension, then a deep neural network is not a universal approximator.

The probabilistic interpretation^[24] derives from the field of machine learning. It features inference,^{[23][7][8][9][12][24]} as well as the optimization concepts of training and testing, related to fitting and generalization, respectively. More specifically, the probabilistic interpretation considers the activation nonlinearity as a cumulative distribution function.^[24] The probabilistic interpretation led to the introduction of dropout as regularizer in neural networks. The probabilistic interpretation was introduced by researchers including Hopfield, Widrow and Narendra and popularized in surveys such as the one by Bishop.^[27]

History

Before 1980

There are two types of artificial neural network (ANN): feedforward neural network (FNN) or multilayer perceptron (MLP) and recurrent neural networks (RNN). RNNs have cycles in their connectivity structure, whereas FNNs do not. In the 1920s, Wilhelm Lenz and Ernst Ising created the Ising model^{[28][29]} which is essentially a non-learning RNN architecture consisting of neuron-like threshold elements. In 1972, Shun'ichi Amari made this architecture adaptive.^{[30][31]} His learning RNN was republished by John Hopfield in 1982.^[32] Other early recurrent neural networks were published by Kaoru Nakano in 1971.^{[33][34]} Already in 1948, Alan Turing produced work on "Intelligent Machinery" that was not published in his lifetime,^[35] containing "ideas related to artificial evolution and learning RNNs".^[31]

Frank Rosenblatt (1958)^[36] proposed the perceptron, an MLP with 3 layers: an input layer, a hidden layer with randomized weights that did not learn, and an output layer. He later published a 1962 book that also introduced variants and computer experiments, including a version with four-layer perceptrons "with adaptive preterminal networks" where the last two layers have learned weights (here he credits H. D. Block and B. W. Knight).^{[37]:section 16} The book cites an earlier network by R. D. Joseph (1960)^[38] "functionally equivalent to a variation of" this four-layer system (the book mentions Joseph over 30 times). Should Joseph therefore be considered the originator of proper adaptive multilayer perceptrons with learning hidden units? Unfortunately, the learning algorithm was not a functional one, and fell into oblivion.

The first working deep learning algorithm was the Group method of data handling, a method to train arbitrarily deep neural networks, published by Alexey Ivakhnenko and Lapa in 1965. They regarded it as a form of polynomial regression,^[39] or a generalization of Rosenblatt's perceptron to handle more complex, nonlinear, and hierarchical relationships.^[40] A 1971 paper described a deep network with eight layers trained by this method,^[41] which is based on layer by layer training through regression analysis. Superfluous hidden units are pruned using a separate validation set. Since the activation functions of the nodes are Kolmogorov-Gabor polynomials, these were also the first deep networks with multiplicative units or "gates".^[31]

The first deep learning multilayer perceptron trained by stochastic gradient descent^[42] was published in 1967 by Shun'ichi Amari.^[43] In computer experiments conducted by Amari's student Saito, a five layer MLP with two modifiable layers learned internal representations to classify non-linearly separable

pattern classes.^[31] Subsequent developments in hardware and hyperparameter tunings have made end-to-end stochastic gradient descent the currently dominant training technique.

In 1969, Kunihiko Fukushima introduced the ReLU (rectified linear unit) activation function.^{[25][31]} The rectifier has become the most popular activation function for deep learning.^[44]

Deep learning architectures for convolutional neural networks (CNNs) with convolutional layers and downsampling layers began with the Neocognitron introduced by Kunihiko Fukushima in 1979, though not trained by backpropagation.^{[45][46]}

Backpropagation is an efficient application of the chain rule derived by Gottfried Wilhelm Leibniz in 1673^[47] to networks of differentiable nodes. The terminology "back-propagating errors" was actually introduced in 1962 by Rosenblatt,^[37] but he did not know how to implement this, although Henry J. Kelley had a continuous precursor of backpropagation in 1960 in the context of control theory.^[48] The modern form of backpropagation was first published in Seppo Linnainmaa's master thesis (1970).^{[49][50][31]} G.M. Ostrovski et al. republished it in 1971.^{[51][52]} Paul Werbos applied backpropagation to neural networks in 1982^[53] (his 1974 PhD thesis, reprinted in a 1994 book,^[54] did not yet describe the algorithm^[52]). In 1986, David E. Rumelhart et al. popularised backpropagation but did not cite the original work.^{[55][56]}

1980s-2000s

The time delay neural network (TDNN) was introduced in 1987 by Alex Waibel to apply CNN to phoneme recognition. It used convolutions, weight sharing, and backpropagation.^{[57][58]} In 1988, Wei Zhang applied a backpropagation-trained CNN to alphabet recognition.^[59] In 1989, Yann LeCun et al. created a CNN called LeNet for recognizing handwritten ZIP codes on mail. Training required 3 days.^[60] In 1990, Wei Zhang implemented a CNN on optical computing hardware.^[61] In 1991, a CNN was applied to medical image object segmentation^[62] and breast cancer detection in mammograms.^[63] LeNet-5 (1998), a 7-level CNN by Yann LeCun et al., that classifies digits, was applied by several banks to recognize hand-written numbers on checks digitized in 32x32 pixel images.^[64]

Recurrent neural networks (RNN)^{[28][30]} were further developed in the 1980s. Recurrence is used for sequence processing, and when a recurrent network is unrolled, it mathematically resembles a deep feedforward layer. Consequently, they have similar properties and issues, and their developments had mutual influences. In RNN, two early influential works were the Jordan network (1986)^[65] and the Elman network (1990),^[66] which applied RNN to study problems in cognitive psychology.

In the 1980s, backpropagation did not work well for deep learning with long credit assignment paths. To overcome this problem, in 1991, Jürgen Schmidhuber proposed a hierarchy of RNNs pre-trained one level at a time by self-supervised learning where each RNN tries to predict its own next input, which is the next unexpected input of the RNN below.^{[67][68]} This "neural history compressor" uses predictive coding to learn internal representations at multiple self-organizing time scales. This can substantially facilitate downstream deep learning. The RNN hierarchy can be *collapsed* into a single RNN, by distilling a higher level *chunker* network into a lower level *automatizer* network.^{[67][68][31]} In 1993, a neural history compressor solved a "Very Deep Learning" task that required more than 1000 subsequent layers in an RNN unfolded in time.^[69] The "P" in ChatGPT refers to such pre-training.

Sepp Hochreiter's diploma thesis (1991)^[70] implemented the neural history compressor,^[67] and identified and analyzed the vanishing gradient problem.^{[70][71]} Hochreiter proposed recurrent residual connections to solve the vanishing gradient problem. This led to the long short-term memory (LSTM), published in 1995.^[72] LSTM can learn "very deep learning" tasks^[9] with long credit assignment paths that require memories of events that happened thousands of discrete time steps before. That LSTM was not yet the modern architecture, which required a "forget gate", introduced in 1999,^[73] which became the standard RNN architecture.

In 1991, Jürgen Schmidhuber also published adversarial neural networks that contest with each other in the form of a zero-sum game, where one network's gain is the other network's loss.^{[74][75]} The first network is a generative model that models a probability distribution over output patterns. The second network learns by gradient descent to predict the reactions of the environment to these patterns. This was called "artificial curiosity". In 2014, this principle was used in generative adversarial networks (GANs).^[76]

During 1985–1995, inspired by statistical mechanics, several architectures and methods were developed by Terry Sejnowski, Peter Dayan, Geoffrey Hinton, etc., including the Boltzmann machine,^[77] restricted Boltzmann machine,^[78] Helmholtz machine,^[79] and the wake-sleep algorithm.^[80] These were designed for unsupervised learning of deep generative models. However, those were more computationally expensive compared to backpropagation. Boltzmann machine learning algorithm, published in 1985, was briefly popular before being eclipsed by the backpropagation algorithm in 1986. (p. 112 ^[81]). A 1988 network became state of the art in protein structure prediction, an early application of deep learning to bioinformatics.^[82]

Both shallow and deep learning (e.g., recurrent nets) of ANNs for speech recognition have been explored for many years.^{[83][84][85]} These methods never outperformed non-uniform internal-handcrafting Gaussian mixture model/Hidden Markov model (GMM-HMM) technology based on generative models of speech trained discriminatively.^[86] Key difficulties have been analyzed, including gradient diminishing^[70] and weak temporal correlation structure in neural predictive models.^{[87][88]} Additional difficulties were the lack of training data and limited computing power.

Most speech recognition researchers moved away from neural nets to pursue generative modeling. An exception was at SRI International in the late 1990s. Funded by the US government's NSA and DARPA, SRI researched in speech and speaker recognition. The speaker recognition team led by Larry Heck reported significant success with deep neural networks in speech processing in the 1998 NIST Speaker Recognition benchmark.^{[89][90]} It was deployed in the Nuance Verifier, representing the first major industrial application of deep learning.^[91]

The principle of elevating "raw" features over hand-crafted optimization was first explored successfully in the architecture of deep autoencoder on the "raw" spectrogram or linear filter-bank features in the late 1990s,^[90] showing its superiority over the Mel-Cepstral features that contain stages of fixed transformation from spectrograms. The raw features of speech, waveforms, later produced excellent larger-scale results.^[92]

2000s

Neural networks entered a lull, and simpler models that use task-specific handcrafted features such as Gabor filters and support vector machines (SVMs) became the preferred choices in the 1990s and 2000s, because of artificial neural networks' computational cost and a lack of understanding of how the brain wires its biological networks.

In 2003, LSTM became competitive with traditional speech recognizers on certain tasks.^[93] In 2006, Alex Graves, Santiago Fernández, Faustino Gomez, and Schmidhuber combined it with connectionist temporal classification (CTC)^[94] in stacks of LSTMs.^[95] In 2009, it became the first RNN to win a pattern recognition contest, in connected handwriting recognition.^{[96][9]}

In 2006, publications by Geoff Hinton, Ruslan Salakhutdinov, Osindero and Teh^{[97][98]} deep belief networks were developed for generative modeling. They are trained by training one restricted Boltzmann machine, then freezing it and training another one on top of the first one, and so on, then optionally fine-tuned using supervised backpropagation.^[97] They could model high-dimensional probability distributions, such as the distribution of MNIST images, but convergence was slow.^{[97][99][100]}

The impact of deep learning in industry began in the early 2000s, when CNNs already processed an estimated 10% to 20% of all the checks written in the US, according to Yann LeCun.^[101] Industrial applications of deep learning to large-scale speech recognition started around 2010.

The 2009 NIPS Workshop on Deep Learning for Speech Recognition was motivated by the limitations of deep generative models of speech, and the possibility that given more capable hardware and large-scale data sets that deep neural nets might become practical. It was believed that pre-training DNNs using generative models of deep belief nets (DBN) would overcome the main difficulties of neural nets. However, it was discovered that replacing pre-training with large amounts of training data for straightforward backpropagation when using DNNs with large, context-dependent output layers produced error rates dramatically lower than then-state-of-the-art Gaussian mixture model (GMM)/Hidden Markov Model (HMM) and also than more-advanced generative model-based systems.^[102] The nature of the recognition errors produced by the two types of systems was characteristically different,^[103] offering technical insights into how to integrate deep learning into the existing highly efficient, run-time speech decoding system deployed by all major speech recognition systems.^{[23][104][105]} Analysis around 2009–2010, contrasting the GMM (and other generative speech models) vs. DNN models, stimulated early industrial investment in deep learning for speech recognition.^[103] That analysis was done with comparable performance (less than 1.5% in error rate) between discriminative DNNs and generative models.^{[102][103][106]} In 2010, researchers extended deep learning from TIMIT to large vocabulary speech recognition, by adopting large output layers of the DNN based on context-dependent HMM states constructed by decision trees.^{[107][108][109][104]}

Deep learning revolution

The deep learning revolution started around CNN- and GPU-based computer vision.

Although CNNs trained by backpropagation had been around for decades and GPU implementations of NNs for years,^[110] including CNNs,^[111] faster implementations of CNNs on GPUs were needed to progress on computer vision. Later, as deep learning becomes widespread, specialized hardware and algorithm optimizations were developed specifically for deep learning.^[112]

A key advance for the deep learning revolution was hardware advances, especially GPU. Some early work dated back to 2004.^{[110][111]} In 2009, Raina, Madhavan, and Andrew Ng reported a 100M deep belief network trained on 30 Nvidia GeForce GTX 280 GPUs, an early demonstration of GPU-based deep learning. They reported up to 70 times faster training.^[113]

In 2011, a CNN named *DanNet*^{[114][115]} by Dan Ciresan, Ueli Meier, Jonathan Masci, Luca Maria Gambardella, and Jürgen Schmidhuber achieved for the first time superhuman performance in a visual pattern recognition contest, outperforming traditional methods by a factor of 3.^[9] It then won more contests.^{[116][117]} They also showed how max-pooling CNNs on GPU improved performance significantly.^[3]

In 2012, Andrew Ng and Jeff Dean created an FNN that learned to recognize higher-level concepts, such as cats, only from watching unlabeled images taken from YouTube videos.^[118]

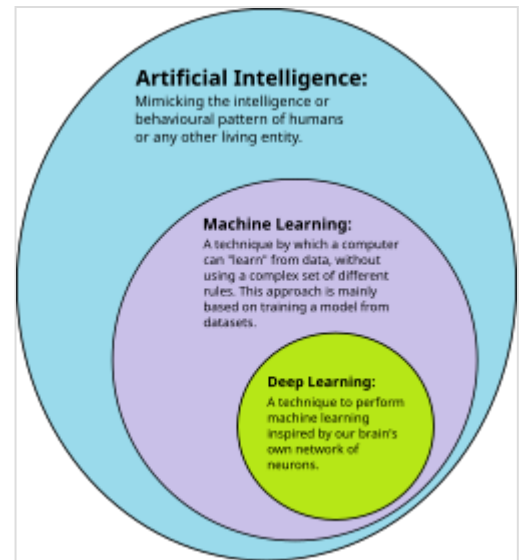
In October 2012, AlexNet by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton^[4] won the large-scale ImageNet competition by a significant margin over shallow machine learning methods. Further incremental improvements included the VGG-16 network by Karen Simonyan and Andrew Zisserman^[119] and Google's Inceptionv3.^[120]

The success in image classification was then extended to the more challenging task of generating descriptions (captions) for images, often as a combination of CNNs and LSTMs.^{[121][122][123]}

In 2014, the state of the art was training "very deep neural network" with 20 to 30 layers.^[124] Stacking too many layers led to a steep reduction in training accuracy,^[125] known as the "degradation" problem.^[126] In 2015, two techniques were developed to train very deep networks: the highway network was published in May 2015, and the residual neural network (ResNet)^[126] in Dec 2015. ResNet behaves like an open-gated Highway Net.

Around the same time, deep learning started impacting the field of art. Early examples included Google DeepDream (2015), and neural style transfer (2015),^[127] both of which were based on pretrained image classification neural networks, such as VGG-19.

Generative adversarial network (GAN) by (Ian Goodfellow et al., 2014)^[128] (based on Jürgen Schmidhuber's principle of artificial curiosity^{[74][76]}) became state of the art in generative modeling during 2014-2018 period. Excellent image quality is achieved by Nvidia's StyleGAN (2018)^[129] based on the Progressive GAN by Tero Karras et al.^[130] Here the GAN generator is grown from small to large scale in a pyramidal fashion. Image generation by GAN reached popular success, and provoked discussions concerning deepfakes.^[131] Diffusion models (2015)^[132] eclipsed GANs in generative modeling since then, with systems such as DALL·E 2 (2022) and Stable Diffusion (2022).



How deep learning is a subset of machine learning and how machine learning is a subset of artificial intelligence (AI)

In 2015, Google's speech recognition improved by 49% by an LSTM-based model, which they made available through Google Voice Search on smartphone.^{[133][134]}

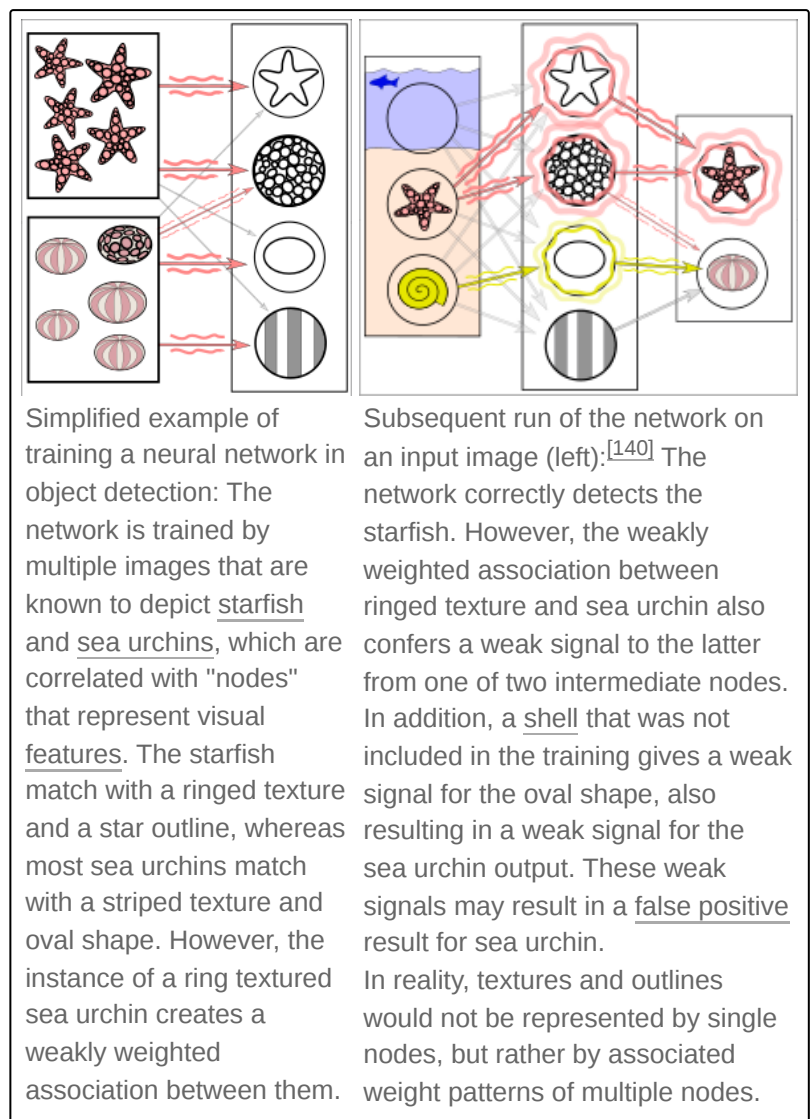
Deep learning is part of state-of-the-art systems in various disciplines, particularly computer vision and automatic speech recognition (ASR). Results on commonly used evaluation sets such as TIMIT (ASR) and MNIST (image classification), as well as a range of large-vocabulary speech recognition tasks have steadily improved.^{[102][135]} Convolutional neural networks were superseded for ASR by LSTM.^{[134][136][137][138]} but are more successful in computer vision.

Yoshua Bengio, Geoffrey Hinton and Yann LeCun were awarded the 2018 Turing Award for "conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing".^[139]

Neural networks

Artificial neural networks (ANNs) or **connectionist systems** are computing systems inspired by the biological neural networks that constitute animal brains. Such systems learn (progressively improve their ability) to do tasks by considering examples, generally without task-specific programming. For example, in image recognition, they might learn to identify images that contain cats by analyzing example images that have been manually labeled as "cat" or "no cat" and using the analytic results to identify cats in other images. They have found most use in applications difficult to express with a traditional computer algorithm using rule-based programming.

An ANN is based on a collection of connected units called artificial neurons, (analogous to biological neurons in a biological brain). Each connection (synapse) between neurons can transmit a signal to another neuron. The receiving (postsynaptic) neuron can process the signal(s) and then signal downstream neurons connected to it. Neurons may have state, generally represented by real numbers, typically between 0 and 1. Neurons and synapses may also have a weight that varies as learning proceeds, which can increase or decrease the strength of the signal that it sends downstream.



Typically, neurons are organized in layers. Different layers may perform different kinds of transformations on their inputs. Signals travel from the first (input), to the last (output) layer, possibly after traversing the layers multiple times.

The original goal of the neural network approach was to solve problems in the same way that a human brain would. Over time, attention focused on matching specific mental abilities, leading to deviations from biology such as backpropagation, or passing information in the reverse direction and adjusting the network to reflect that information.

Neural networks have been used on a variety of tasks, including computer vision, speech recognition, machine translation, social network filtering, playing board and video games and medical diagnosis.

As of 2017, neural networks typically have a few thousand to a few million units and millions of connections. Despite this number being several order of magnitude less than the number of neurons on a human brain, these networks can perform many tasks at a level beyond that of humans (e.g., recognizing faces, or playing "Go"^[141]).

Deep neural networks

A **deep neural network (DNN)** is an artificial neural network with multiple layers between the input and output layers.^{[7][9]} There are different types of neural networks but they always consist of the same components: neurons, synapses, weights, biases, and functions.^[142] These components as a whole function in a way that mimics functions of the human brain, and can be trained like any other ML algorithm.

For example, a DNN that is trained to recognize dog breeds will go over the given image and calculate the probability that the dog in the image is a certain breed. The user can review the results and select which probabilities the network should display (above a certain threshold, etc.) and return the proposed label. Each mathematical manipulation as such is considered a layer,^[143] and complex DNN have many layers, hence the name "deep" networks.

DNNs can model complex non-linear relationships. DNN architectures generate compositional models where the object is expressed as a layered composition of primitives.^[144] The extra layers enable composition of features from lower layers, potentially modeling complex data with fewer units than a similarly performing shallow network.^[7] For instance, it was proved that sparse multivariate polynomials are exponentially easier to approximate with DNNs than with shallow networks.^[145]

Deep architectures include many variants of a few basic approaches. Each architecture has found success in specific domains. It is not always possible to compare the performance of multiple architectures, unless they have been evaluated on the same data sets.^[143]

DNNs are typically feedforward networks in which data flows from the input layer to the output layer without looping back. At first, the DNN creates a map of virtual neurons and assigns random numerical values, or "weights", to connections between them. The weights and inputs are multiplied and return an output between 0 and 1. If the network did not accurately recognize a particular pattern, an algorithm would adjust the weights.^[146] That way the algorithm can make certain parameters more influential, until it determines the correct mathematical manipulation to fully process the data.

Recurrent neural networks, in which data can flow in any direction, are used for applications such as language modeling.^{[147][148][149][150][151]} Long short-term memory is particularly effective for this use.^{[152][153]}

Convolutional neural networks (CNNs) are used in computer vision.^[154] CNNs also have been applied to acoustic modeling for automatic speech recognition (ASR).^[155]

Challenges

As with ANNs, many issues can arise with naively trained DNNs. Two common issues are overfitting and computation time.

DNNs are prone to overfitting because of the added layers of abstraction, which allow them to model rare dependencies in the training data. Regularization methods such as Ivakhnenko's unit pruning^[41] or weight decay (ℓ_2 -regularization) or sparsity (ℓ_1 -regularization) can be applied during training to combat overfitting.^[156] Alternatively dropout regularization randomly omits units from the hidden layers during training. This helps to exclude rare dependencies.^[157] Another interesting recent development is research into models of just enough complexity through an estimation of the intrinsic complexity of the task being modelled. This approach has been successfully applied for multivariate time series prediction tasks such as traffic prediction.^[158] Finally, data can be augmented via methods such as cropping and rotating such that smaller training sets can be increased in size to reduce the chances of overfitting.^[159]

DNNs must consider many training parameters, such as the size (number of layers and number of units per layer), the learning rate, and initial weights. Sweeping through the parameter space for optimal parameters may not be feasible due to the cost in time and computational resources. Various tricks, such as batching (computing the gradient on several training examples at once rather than individual examples)^[160] speed up computation. Large processing capabilities of many-core architectures (such as GPUs or the Intel Xeon Phi) have produced significant speedups in training, because of the suitability of such processing architectures for the matrix and vector computations.^{[161][162]}

Alternatively, engineers may look for other types of neural networks with more straightforward and convergent training algorithms. CMAC (cerebellar model articulation controller) is one such kind of neural network. It doesn't require learning rates or randomized initial weights. The training process can be guaranteed to converge in one step with a new batch of data, and the computational complexity of the training algorithm is linear with respect to the number of neurons involved.^{[163][164]}

Hardware

Since the 2010s, advances in both machine learning algorithms and computer hardware have led to more efficient methods for training deep neural networks that contain many layers of non-linear hidden units and a very large output layer.^[165] By 2019, graphics processing units (GPUs), often with AI-specific enhancements, had displaced CPUs as the dominant method for training large-scale commercial cloud AI.^[166] OpenAI estimated the hardware computation used in the largest deep learning projects from AlexNet (2012) to AlphaZero (2017) and found a 300,000-fold increase in the amount of computation required, with a doubling-time trendline of 3.4 months.^{[167][168]}

Special electronic circuits called deep learning processors were designed to speed up deep learning algorithms. Deep learning processors include neural processing units (NPUs) in Huawei cellphones^[169] and cloud computing servers such as tensor processing units (TPU) in the Google Cloud Platform.^[170] Cerebras Systems has also built a dedicated system to handle large deep learning models, the CS-2, based on the largest processor in the industry, the second-generation Wafer Scale Engine (WSE-2).^{[171][172]}

Atomically thin semiconductors are considered promising for energy-efficient deep learning hardware where the same basic device structure is used for both logic operations and data storage. In 2020, Marega et al. published experiments with a large-area active channel material for developing logic-in-memory devices and circuits based on floating-gate field-effect transistors (FGFETs).^[173]

In 2021, J. Feldmann et al. proposed an integrated photonic hardware accelerator for parallel convolutional processing.^[174] The authors identify two key advantages of integrated photonics over its electronic counterparts: (1) massively parallel data transfer through wavelength division multiplexing in conjunction with frequency combs, and (2) extremely high data modulation speeds.^[174] Their system can execute trillions of multiply-accumulate operations per second, indicating the potential of integrated photonics in data-heavy AI applications.^[174]

Applications

Automatic speech recognition

Large-scale automatic speech recognition is the first and most convincing successful case of deep learning. LSTM RNNs can learn "Very Deep Learning" tasks^[9] that involve multi-second intervals containing speech events separated by thousands of discrete time steps, where one time step corresponds to about 10 ms. LSTM with forget gates^[153] is competitive with traditional speech recognizers on certain tasks.^[93]

The initial success in speech recognition was based on small-scale recognition tasks based on TIMIT. The data set contains 630 speakers from eight major dialects of American English, where each speaker reads 10 sentences.^[175] Its small size lets many configurations be tried. More importantly, the TIMIT task concerns phone-sequence recognition, which, unlike word-sequence recognition, allows weak phone bigram language models. This lets the strength of the acoustic modeling aspects of speech recognition be more easily analyzed. The error rates listed below, including these early results and measured as percent phone error rates (PER), have been summarized since 1991.

Method	Percent phone error rate (PER) (%)
Randomly Initialized RNN ^[176]	26.1
Bayesian Triphone GMM-HMM	25.6
Hidden Trajectory (Generative) Model	24.8
Monophone Randomly Initialized DNN	23.4
Monophone DBN-DNN	22.4
Triphone GMM-HMM with BMMI Training	21.7
Monophone DBN-DNN on fbank	20.7
Convolutional DNN ^[177]	20.0
Convolutional DNN w. Heterogeneous Pooling	18.7
Ensemble DNN/CNN/RNN ^[178]	18.3
Bidirectional LSTM	17.8
Hierarchical Convolutional Deep Maxout Network ^[179]	16.5

The debut of DNNs for speaker recognition in the late 1990s and speech recognition around 2009-2011 and of LSTM around 2003–2007, accelerated progress in eight major areas:^{[23][106][104]}

- Scale-up/out and accelerated DNN training and decoding
- Sequence discriminative training
- Feature processing by deep models with solid understanding of the underlying mechanisms
- Adaptation of DNNs and related deep models
- Multi-task and transfer learning by DNNs and related deep models
- CNNs and how to design them to best exploit domain knowledge of speech
- RNN and its rich LSTM variants
- Other types of deep models including tensor-based models and integrated deep generative/discriminative models.

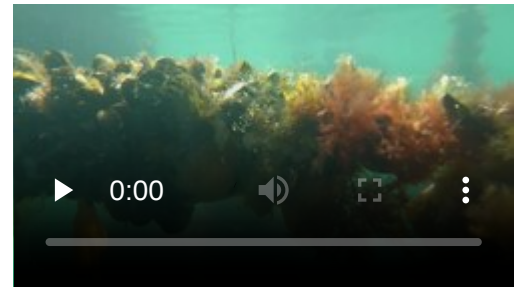
More recent speech recognition models use Transformers or Temporal Convolution Networks with significant success and widespread applications.^{[180][181][182]} All major commercial speech recognition systems (e.g., Microsoft Cortana, Xbox, Skype Translator, Amazon Alexa, Google Now, Apple Siri, Baidu and iFlyTek voice search, and a range of Nuance speech products, etc.) are based on deep learning.^{[23][183][184]}

Image recognition

A common evaluation set for image classification is the MNIST database data set. MNIST is composed of handwritten digits and includes 60,000 training examples and 10,000 test examples. As with TIMIT, its small size lets users test multiple configurations. A comprehensive list of results on this set is available.^[185]

Deep learning-based image recognition has become "superhuman", producing more accurate results than human contestants. This first occurred in 2011 in recognition of traffic signs, and in 2014, with recognition of human faces.^{[186][187]}

Deep learning-trained vehicles now interpret 360° camera views.^[188] Another example is Facial Dysmorphology Novel Analysis (FDNA) used to analyze cases of human malformation connected to a large database of genetic syndromes.



Richard Green explains how deep learning is used with a remotely operated vehicle in mussel aquaculture.

Visual art processing

Closely related to the progress that has been made in image recognition is the increasing application of deep learning techniques to various visual art tasks. DNNs have proven themselves capable, for example, of

- identifying the style period of a given painting^{[189][190]}
- Neural Style Transfer – capturing the style of a given artwork and applying it in a visually pleasing manner to an arbitrary photograph or video^{[189][190]}
- generating striking imagery based on random visual input fields.^{[189][190]}



Visual art processing of Jimmy Wales in France, with the style of Munch's "The Scream" applied using neural style transfer

Natural language processing

Neural networks have been used for implementing language models since the early 2000s.^[147] LSTM helped to improve machine translation and language modeling.^{[148][149][150]}

Other key techniques in this field are negative sampling^[191] and word embedding. Word embedding, such as word2vec, can be thought of as a representational layer in a deep learning architecture that transforms an atomic word into a positional representation of the word relative to other words in the dataset; the position is represented as a point in a vector space. Using word embedding as an RNN input layer allows the network to parse sentences and phrases using an effective compositional vector grammar. A compositional vector grammar can be thought of as probabilistic context free grammar (PCFG) implemented by an RNN.^[192] Recursive auto-encoders built atop word embeddings can assess sentence similarity and detect paraphrasing.^[192] Deep neural architectures provide the best results for constituency parsing,^[193] sentiment analysis,^[194] information retrieval,^{[195][196]} spoken language understanding,^[197] machine translation,^{[148][198]} contextual entity linking,^[198] writing style recognition,^[199] named-entity recognition (token classification),^[200] text classification, and others.^[201]

Recent developments generalize word embedding to sentence embedding.

Google Translate (GT) uses a large end-to-end long short-term memory (LSTM) network.^{[202][203][204][205]} Google Neural Machine Translation (GNMT) uses an example-based machine translation method in which the system "learns from millions of examples".^[203] It translates "whole sentences at a time, rather than pieces". Google Translate supports over one hundred languages.^[203] The network encodes the "semantics of the sentence rather than simply memorizing phrase-to-phrase translations".^{[203][206]} GT uses English as an intermediate between most language pairs.^[206]

Drug discovery and toxicology

A large percentage of candidate drugs fail to win regulatory approval. These failures are caused by insufficient efficacy (on-target effect), undesired interactions (off-target effects), or unanticipated toxic effects.^{[207][208]} Research has explored use of deep learning to predict the biomolecular targets,^{[209][210]} off-targets, and toxic effects of environmental chemicals in nutrients, household products and drugs.^{[211][212][213]}

AtomNet is a deep learning system for structure-based rational drug design.^[214] AtomNet was used to predict novel candidate biomolecules for disease targets such as the Ebola virus^[215] and multiple sclerosis.^{[216][215]}

In 2017 graph neural networks were used for the first time to predict various properties of molecules in a large toxicology data set.^[217] In 2019, generative neural networks were used to produce molecules that were validated experimentally all the way into mice.^{[218][219]}

Recommendation systems

Recommendation systems have used deep learning to extract meaningful features for a latent factor model for content-based music and journal recommendations.^{[220][221]} Multi-view deep learning has been applied for learning user preferences from multiple domains.^[222] The model uses a hybrid collaborative and content-based approach and enhances recommendations in multiple tasks.

Bioinformatics

An autoencoder ANN was used in bioinformatics, to predict gene ontology annotations and gene-function relationships.^[223]

In medical informatics, deep learning was used to predict sleep quality based on data from wearables^[224] and predictions of health complications from electronic health record data.^[225]

Deep neural networks have shown unparalleled performance in predicting protein structure, according to the sequence of the amino acids that make it up. In 2020, AlphaFold, a deep-learning based system, achieved a level of accuracy significantly higher than all previous computational methods.^{[226][227]}

Deep Neural Network Estimations

Deep neural networks can be used to estimate the entropy of a stochastic process through an arrangement called a Neural Joint Entropy Estimator (NJEE).^[228] Such an estimation provides insights on the effects of input random variables on an independent random variable. Practically, the DNN is trained as a classifier that maps an input vector or matrix X to an output probability distribution over the possible classes of random variable Y, given input X. For example, in image classification tasks, the NJEE maps a vector of pixels' color values to probabilities over possible image classes. In practice, the probability distribution of Y is obtained by a Softmax layer with number of nodes that is equal to the alphabet size of Y. NJEE uses continuously differentiable activation functions, such that the conditions for the universal approximation theorem holds. It is shown that this method provides a strongly consistent estimator and outperforms other methods in cases of large alphabet sizes.^[228]

Medical image analysis

Deep learning has been shown to produce competitive results in medical applications such as cancer cell classification, lesion detection, organ segmentation and image enhancement.^{[229][230]} Modern deep learning tools demonstrate the high accuracy of detecting various diseases and the helpfulness of their use by specialists to improve the diagnosis efficiency.^{[231][232]}

Mobile advertising

Finding the appropriate mobile audience for mobile advertising is always challenging, since many data points must be considered and analyzed before a target segment can be created and used in ad serving by any ad server.^[233] Deep learning has been used to interpret large, many-dimensioned advertising datasets. Many data points are collected during the request/serve/click internet advertising cycle. This information can form the basis of machine learning to improve ad selection.

Image restoration

Deep learning has been successfully applied to inverse problems such as denoising, super-resolution, inpainting, and film colorization.^[234] These applications include learning methods such as "Shrinkage Fields for Effective Image Restoration"^[235] which trains on an image dataset, and Deep Image Prior, which trains on the image that needs restoration.

Financial fraud detection

Deep learning is being applied to financial fraud detection, tax evasion detection,^[236] and anti-money laundering.^[237]

Materials science

In November 2023, researchers at Google DeepMind and Lawrence Berkeley National Laboratory announced that they had developed an AI system known as GNoME. This system has contributed to materials science by discovering over 2 million new materials within a relatively short timeframe. GNoME employs deep learning techniques to efficiently explore potential material structures, achieving a significant increase in the identification of stable inorganic crystal structures. The system's predictions were validated through autonomous robotic experiments, demonstrating a noteworthy success rate of 71%. The data of newly discovered materials is publicly available through the Materials Project database, offering researchers the opportunity to identify materials with desired properties for various applications. This development has implications for the future of scientific discovery and the integration of AI in material science research, potentially expediting material innovation and reducing costs in product development. The use of AI and deep learning suggests the possibility of minimizing or eliminating manual lab experiments and allowing scientists to focus more on the design and analysis of unique compounds.^{[238][239][240]}

Military

The United States Department of Defense applied deep learning to train robots in new tasks through observation.^[241]

Partial differential equations

Physics informed neural networks have been used to solve partial differential equations in both forward and inverse problems in a data driven manner.^[242] One example is the reconstructing fluid flow governed by the Navier-Stokes equations. Using physics informed neural networks does not require the often expensive mesh generation that conventional CFD methods rely on.^{[243][244]} It is evident that geometric and physical constraints have a synergistic effect on neural PDE surrogates, thereby enhancing their efficacy in predicting stable and super long rollouts.^[245]

Deep backward stochastic differential equation method

Deep backward stochastic differential equation method is a numerical method that combines deep learning with Backward stochastic differential equation (BSDE). This method is particularly useful for solving high-dimensional problems in financial mathematics. By leveraging the powerful function approximation capabilities of deep neural networks, deep BSDE addresses the computational challenges faced by traditional numerical methods in high-dimensional settings. Specifically, traditional methods like finite difference methods or Monte Carlo simulations often struggle with the curse of dimensionality, where computational cost increases exponentially with the number of dimensions. Deep BSDE methods, however, employ deep neural networks to approximate solutions of high-dimensional partial differential equations (PDEs), effectively reducing the computational burden.^[246]

In addition, the integration of Physics-informed neural networks (PINNs) into the deep BSDE framework enhances its capability by embedding the underlying physical laws directly into the neural network architecture. This ensures that the solutions not only fit the data but also adhere to the governing stochastic differential equations. PINNs leverage the power of deep learning while respecting the constraints imposed by the physical models, resulting in more accurate and reliable solutions for financial mathematics problems.

Image reconstruction

Image reconstruction is the reconstruction of the underlying images from the image-related measurements. Several works showed the better and superior performance of the deep learning methods compared to analytical methods for various applications, e.g., spectral imaging ^[247] and ultrasound imaging.^[248]

Weather prediction

Traditional weather prediction systems solve a very complex system of partial differential equations. GraphCast is a deep learning based model, trained on a long history of weather data to predict how weather patterns change over time. It is able to predict weather conditions for up to 10 days globally, at a very detailed level, and in under a minute, with precision similar to state of the art systems.^{[249][250]}

Epigenetic clock

An epigenetic clock is a biochemical test that can be used to measure age. Galkin et al. used deep neural networks to train an epigenetic aging clock of unprecedented accuracy using >6,000 blood samples.^[251] The clock uses information from 1000 CpG sites and predicts people with certain conditions older than

healthy controls: IBD, frontotemporal dementia, ovarian cancer, obesity. The aging clock was planned to be released for public use in 2021 by an Insilico Medicine spinoff company Deep Longevity.

Relation to human cognitive and brain development

Deep learning is closely related to a class of theories of brain development (specifically, neocortical development) proposed by cognitive neuroscientists in the early 1990s.^{[252][253][254][255]} These developmental theories were instantiated in computational models, making them predecessors of deep learning systems. These developmental models share the property that various proposed learning dynamics in the brain (e.g., a wave of nerve growth factor) support the self-organization somewhat analogous to the neural networks utilized in deep learning models. Like the neocortex, neural networks employ a hierarchy of layered filters in which each layer considers information from a prior layer (or the operating environment), and then passes its output (and possibly the original input), to other layers. This process yields a self-organizing stack of transducers, well-tuned to their operating environment. A 1995 description stated, "...the infant's brain seems to organize itself under the influence of waves of so-called trophic-factors ... different regions of the brain become connected sequentially, with one layer of tissue maturing before another and so on until the whole brain is mature".^[256]

A variety of approaches have been used to investigate the plausibility of deep learning models from a neurobiological perspective. On the one hand, several variants of the backpropagation algorithm have been proposed in order to increase its processing realism.^{[257][258]} Other researchers have argued that unsupervised forms of deep learning, such as those based on hierarchical generative models and deep belief networks, may be closer to biological reality.^{[259][260]} In this respect, generative neural network models have been related to neurobiological evidence about sampling-based processing in the cerebral cortex.^[261]

Although a systematic comparison between the human brain organization and the neuronal encoding in deep networks has not yet been established, several analogies have been reported. For example, the computations performed by deep learning units could be similar to those of actual neurons^[262] and neural populations.^[263] Similarly, the representations developed by deep learning models are similar to those measured in the primate visual system^[264] both at the single-unit^[265] and at the population^[266] levels.

Commercial activity

Facebook's AI lab performs tasks such as automatically tagging uploaded pictures with the names of the people in them.^[267]

Google's DeepMind Technologies developed a system capable of learning how to play Atari video games using only pixels as data input. In 2015 they demonstrated their AlphaGo system, which learned the game of Go well enough to beat a professional Go player.^{[268][269][270]} Google Translate uses a neural network to translate between more than 100 languages.

In 2017, Covariant.ai was launched, which focuses on integrating deep learning into factories.^[271]

As of 2008,^[272] researchers at The University of Texas at Austin (UT) developed a machine learning framework called Training an Agent Manually via Evaluative Reinforcement, or TAMER, which proposed new methods for robots or computer programs to learn how to perform tasks by interacting with a human instructor.^[241] First developed as TAMER, a new algorithm called Deep TAMER was later introduced in 2018 during a collaboration between U.S. Army Research Laboratory (ARL) and UT researchers. Deep TAMER used deep learning to provide a robot with the ability to learn new tasks through observation.^[241] Using Deep TAMER, a robot learned a task with a human trainer, watching video streams or observing a human perform a task in-person. The robot later practiced the task with the help of some coaching from the trainer, who provided feedback such as "good job" and "bad job".^[273]

Criticism and comment

Deep learning has attracted both criticism and comment, in some cases from outside the field of computer science.

Theory

A main criticism concerns the lack of theory surrounding some methods.^[274] Learning in the most common deep architectures is implemented using well-understood gradient descent. However, the theory surrounding other algorithms, such as contrastive divergence is less clear. (e.g., Does it converge? If so, how fast? What is it approximating?) Deep learning methods are often looked at as a black box, with most confirmations done empirically, rather than theoretically.^[275]

In further reference to the idea that artistic sensitivity might be inherent in relatively low levels of the cognitive hierarchy, a published series of graphic representations of the internal states of deep (20-30 layers) neural networks attempting to discern within essentially random data the images on which they were trained^[276] demonstrate a visual appeal: the original research notice received well over 1,000 comments, and was the subject of what was for a time the most frequently accessed article on The Guardian's^[277] website.

With the support of Innovation Diffusion Theory (IDT), a study analyzed the diffusion of Deep Learning^[278] in BRICS and OECD countries using data from Google Trends.

Errors

Some deep learning architectures display problematic behaviors,^[279] such as confidently classifying unrecognizable images as belonging to a familiar category of ordinary images (2014)^[280] and misclassifying minuscule perturbations of correctly classified images (2013).^[281] Goertzel hypothesized that these behaviors are due to limitations in their internal representations and that these limitations would inhibit integration into heterogeneous multi-component artificial general intelligence (AGI) architectures.^[279] These issues may possibly be addressed by deep learning architectures that internally form states homologous to image-grammar^[282] decompositions of observed entities and events.^[279] Learning a grammar (visual or linguistic) from training data would be equivalent to restricting the system to commonsense reasoning that operates on concepts in terms of grammatical production rules and is a basic goal of both human language acquisition^[283] and artificial intelligence (AI).^[284]

Cyber threat

As deep learning moves from the lab into the world, research and experience show that artificial neural networks are vulnerable to hacks and deception.^[285] By identifying patterns that these systems use to function, attackers can modify inputs to ANNs in such a way that the ANN finds a match that human observers would not recognize. For example, an attacker can make subtle changes to an image such that the ANN finds a match even though the image looks to a human nothing like the search target. Such manipulation is termed an "adversarial attack".^[286]

In 2016 researchers used one ANN to doctor images in trial and error fashion, identify another's focal points, and thereby generate images that deceived it. The modified images looked no different to human eyes. Another group showed that printouts of doctored images then photographed successfully tricked an image classification system.^[287] One defense is reverse image search, in which a possible fake image is submitted to a site such as TinEye that can then find other instances of it. A refinement is to search using only parts of the image, to identify images from which that piece may have been taken.^[288]

Another group showed that certain psychedelic spectacles could fool a facial recognition system into thinking ordinary people were celebrities, potentially allowing one person to impersonate another. In 2017 researchers added stickers to stop signs and caused an ANN to misclassify them.^[287]

ANNs can however be further trained to detect attempts at deception, potentially leading attackers and defenders into an arms race similar to the kind that already defines the malware defense industry. ANNs have been trained to defeat ANN-based anti-malware software by repeatedly attacking a defense with malware that was continually altered by a genetic algorithm until it tricked the anti-malware while retaining its ability to damage the target.^[287]

In 2016, another group demonstrated that certain sounds could make the Google Now voice command system open a particular web address, and hypothesized that this could "serve as a stepping stone for further attacks (e.g., opening a web page hosting drive-by malware)".^[287]

In "data poisoning", false data is continually smuggled into a machine learning system's training set to prevent it from achieving mastery.^[287]

Data collection ethics

The deep learning systems that are trained using supervised learning often rely on data that is created or annotated by humans, or both.^[289] It has been argued that not only low-paid clickwork (such as on Amazon Mechanical Turk) is regularly deployed for this purpose, but also implicit forms of human microwork that are often not recognized as such.^[290] The philosopher Rainer Mùhlhoff distinguishes five types of "machinic capture" of human microwork to generate training data: (1) gamification (the embedding of annotation or computation tasks in the flow of a game), (2) "trapping and tracking" (e.g. CAPTCHAs for image recognition or click-tracking on Google search results pages), (3) exploitation of social motivations (e.g. tagging faces on Facebook to obtain labeled facial images), (4) information mining (e.g. by leveraging quantified-self devices such as activity trackers) and (5) clickwork.^[290]

Machine learning

Machine learning (**ML**) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without being explicitly programmed.^[1] Advances in the field of deep learning have allowed neural networks, a class of statistical algorithms, to surpass many previous machine learning approaches in performance.

Statistics and mathematical optimisation methods compose the foundations of machine learning. Data mining is a related field of study, focusing on exploratory data analysis (EDA) through unsupervised learning.^{[3][4]}

From a theoretical viewpoint, probably approximately correct learning provides a mathematical and statistical framework for describing machine learning. Most traditional machine learning and deep learning algorithms can be described as empirical risk minimisation under this framework.

History

The term *machine learning* was coined in 1959 by Arthur Samuel, an IBM employee and pioneer in the field of computer gaming and artificial intelligence.^{[5][6]} The synonym *self-teaching computers* was also used during this time period.^{[7][8]}

The earliest machine learning program was introduced in the 1950s, when Samuel invented a computer program that calculated the chance of winning in checkers for each side, but the history of machine learning is rooted in decades of efforts to study human cognitive processes.^[9] In 1949, Canadian psychologist Donald Hebb published the book *The Organization of Behavior*, in which he introduced a theoretical neural structure formed by certain interactions among nerve cells.^[10] The Hebbian theory of neuron interaction set the groundwork for how many machine learning algorithms work, with connected artificial neurons changing the strength of their connections based on data.^[9] Other researchers who have studied human cognitive systems contributed to the modern machine learning technologies as well, including Walter Pitts and Warren McCulloch, who proposed the first mathematical model of neural networks including algorithms that mirror human thought processes.^[9]

By the early 1960s, an experimental "learning machine" with punched tape memory, called Cybertron, had been developed by Raytheon Company to analyse sonar signals, electrocardiograms, and speech patterns using rudimentary reinforcement learning. It was repetitively "trained" by a human operator/teacher to recognise patterns and equipped with a "goof" button to cause it to reevaluate incorrect decisions.^[11] A representative book on research into machine learning during the 1960s was Nils Nilsson's book "Learning Machines", dealing mostly with machine learning for pattern classification.^[12] Interest related to pattern recognition continued into the 1970s, as described by Duda and Hart in 1973.^[13] In 1981, a report was given on using teaching strategies so that an artificial neural network learns to recognise 40 characters (26 letters, 10 digits, and 4 special symbols) from a computer terminal.^[14]

Tom M. Mitchell provided a widely quoted, more formal definition of the algorithms studied in the machine learning field: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E ."^[15] This definition of the tasks in which machine learning is concerned is fundamentally operational rather than defining the field in cognitive terms. This follows Alan Turing's proposal in his paper "Computing Machinery and Intelligence", in which the question, "Can machines think?", is replaced by asking whether machines can convincingly imitate a human in its responses to human-posed questions.^{[16][17]}

In 2014 Ian Goodfellow and others introduced generative adversarial networks (GANs) which could produce realistic synthetic data.^[18] By 2016 AlphaGo had won against top human players in Go using reinforcement learning techniques.^[19]

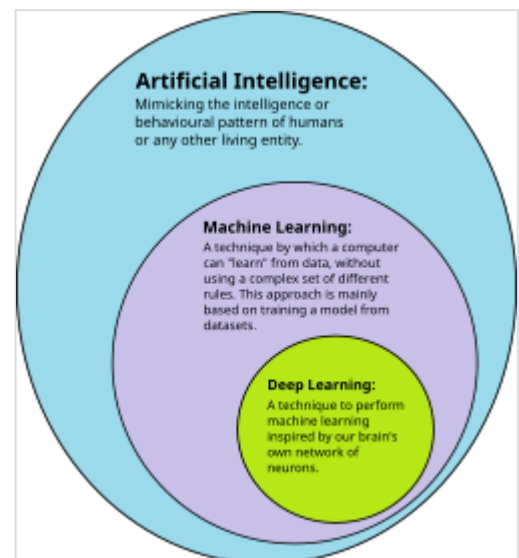
Relationships to other fields

Artificial intelligence

As a scientific endeavour, machine learning grew out of the quest for artificial intelligence (AI). In the early days of AI as an academic discipline, some researchers were interested in having machines learn from data. They attempted to approach the problem with various symbolic methods, as well as what were then termed "neural networks"; these were mostly perceptrons and other models that were later found to be reinventions of the generalised linear models of statistics.^[21] Probabilistic reasoning was also employed, especially in automated medical diagnosis.^{[22]:488}

However, an increasing emphasis on the logical, knowledge-based approach caused a rift between AI and machine learning. Probabilistic systems were plagued by theoretical and practical problems of data acquisition and representation.^{[22]:488} By 1980, expert systems had come to dominate AI, and statistics was out of favour.^[23] Work on symbolic/knowledge-based learning continued within AI, leading to inductive logic programming (ILP), but the more statistical line of research was now outside the field of AI proper, in pattern recognition and information retrieval.^{[22]:708–710,755} Neural network research was abandoned by AI and computer science around the same time. This subfield, termed "connectionism", was continued by researchers from other disciplines, including John Hopfield, David Rumelhart, and Geoffrey Hinton. Their main success came in the mid-1980s with the reinvention of backpropagation.^{[22]:25}

Machine learning (ML), reorganised and recognised as its own field, started to flourish in the 1990s. The field changed its goal from achieving artificial intelligence to tackling solvable problems of a practical nature. It shifted focus away from the symbolic approaches it had inherited from AI, and toward methods and models borrowed from statistics, fuzzy logic, and probability theory.^[23]



Deep learning is a subset of machine learning, which is itself a subset of artificial intelligence.^[20]

Data compression

There is a close connection between machine learning and compression. A system that predicts the posterior probabilities of a sequence given its entire history can be used for optimal data compression (by using arithmetic coding on the output distribution). Conversely, an optimal compressor can be used for prediction (by finding the symbol that compresses best, given the previous history). This equivalence has been used as a justification for using data compression as a benchmark for "general intelligence".^{[24][25][26]}

An alternative view can show compression algorithms implicitly map strings into implicit feature space vectors, and compression-based similarity measures compute similarity within these feature spaces. For each compressor $C(\cdot)$ we define an associated vector space $\mathcal{X}$, such that $C(\cdot)$ maps an input string x , corresponding to the vector norm $\|\cdot\|$. An exhaustive examination of the feature spaces underlying all compression algorithms is precluded by space; instead, feature vectors chooses to examine three representative lossless compression methods, LZW, LZ77, and PPM.^[27]

According to AIXI theory, a connection more directly explained in Hutter Prize, the best possible compression of x is the smallest possible software that generates x . For example, in that model, a zip file's compressed size includes both the zip file and the unzipping software, since you can not unzip it without both, but there may be an even smaller combined form.

Examples of AI-powered audio/video compression software include NVIDIA Maxine, AIVC.^[28] Examples of software that can perform AI-powered image compression include OpenCV, TensorFlow, MATLAB's Image Processing Toolbox (IPT) and High-Fidelity Generative Image Compression.^[29]

In unsupervised machine learning, k-means clustering can be utilized to compress data by grouping similar data points into clusters. This technique simplifies handling extensive datasets that lack predefined labels and finds widespread use in fields such as image compression.^[30]

Data compression aims to reduce the size of data files, enhancing storage efficiency and speeding up data transmission. K-means clustering, an unsupervised machine learning algorithm, is employed to partition a dataset into a specified number of clusters, k , each represented by the centroid of its points. This process condenses extensive datasets into a more compact set of representative points. Particularly beneficial in image and signal processing, k-means clustering aids in data reduction by replacing groups of data points with their centroids, thereby preserving the core information of the original data while significantly decreasing the required storage space.^[31]

Large language models (LLMs) are also efficient lossless data compressors on some data sets, as demonstrated by DeepMind's research with the Chinchilla 70B model. Developed by DeepMind, Chinchilla 70B effectively compressed data, outperforming conventional methods such as Portable Network Graphics (PNG) for images and Free Lossless Audio Codec (FLAC) for audio. It achieved compression of image and audio data to 43.4% and 16.4% of their original sizes, respectively. There is, however, some reason to be concerned that the data set used for testing overlaps the LLM training data set, making it possible that the Chinchilla 70B model is only an efficient compression tool on data it has already been trained on.^{[32][33]}

Data mining

Machine learning and data mining often employ the same methods and overlap significantly, but while machine learning focuses on prediction based on *known* properties learned from the training data, data mining focuses on the discovery of previously *unknown* properties in the data (this is the analysis step of knowledge discovery in databases). Data mining uses many machine learning methods, but with different goals; on the other hand, machine learning also employs data mining methods as "unsupervised learning" or as a preprocessing step to improve learner accuracy. Much of the confusion between these two research communities comes from the basic assumptions they work with: in machine learning, performance is usually evaluated with respect to the ability to *reproduce known* knowledge, while in knowledge discovery and data mining (KDD) the key task is the discovery of previously *unknown* knowledge. Evaluated with respect to known knowledge, an uninformed (unsupervised) method will easily be outperformed by other supervised methods, while in a typical KDD task, supervised methods cannot be used due to the unavailability of training data.

Machine learning also has intimate ties to optimization: Many learning problems are formulated as minimisation of some loss function on a training set of examples. Loss functions express the discrepancy between the predictions of the model being trained and the actual problem instances (for example, in classification, one wants to assign a label to instances, and models are trained to correctly predict the preassigned labels of a set of examples).^[34]

Generalization

Characterizing the generalisation of various learning algorithms is an active topic of current research, especially for deep learning algorithms.

Statistics

Machine learning and statistics are closely related fields in terms of methods, but distinct in their principal goal: statistics draws population inferences from a sample, while machine learning finds generalisable predictive patterns.^[35]

Conventional statistical analyses require the a priori selection of a model most suitable for the study data set. In addition, only significant or theoretically relevant variables based on previous experience are included for analysis. In contrast, machine learning is not built on a pre-structured model; rather, the data shape the model by detecting underlying patterns. The more variables (input) used to train the model, the more accurate the ultimate model will be.^[36]

Leo Breiman distinguished two statistical modelling paradigms: the data model and the algorithmic model,^[37] wherein "algorithmic model" means more or less the machine learning algorithms like Random forest.

Some statisticians have adopted methods from machine learning, producing the field of *statistical learning*.^[38]

Statistical physics

Analytical and computational techniques derived from deep-rooted physics of disordered systems can be extended to large-scale problems, including machine learning, e.g., to analyse the weight space of deep neural networks.^[39] Statistical physics is thus finding applications in the area of medical diagnostics.^[40]

Theory

A core objective of a learner is to generalise from its experience.^{[2][41]} Generalisation in this context is the ability of a learning machine to perform accurately on new, unseen examples/tasks after having experienced a learning data set. The training examples come from some generally unknown probability distribution (considered representative of the space of occurrences) and the learner has to build a general model about this space that enables it to produce sufficiently accurate predictions in new cases.

The computational analysis of machine learning algorithms and their performance is a branch of theoretical computer science known as computational learning theory via the probably approximately correct learning model. Because training sets are finite and the future is uncertain, learning theory usually does not yield guarantees of the performance of algorithms. Instead, probabilistic bounds on the performance are quite common. The bias–variance decomposition is one way to quantify generalisation error.

For the best performance in the context of generalisation, the complexity of the hypothesis should match the complexity of the function underlying the data. If the hypothesis is less complex than the function, then the model has underfitted the data. If the complexity of the model is increased in response, then the training error decreases. But if the hypothesis is too complex, then the model is subject to overfitting and generalisation will be poorer.^[42]

In addition to performance bounds, learning theorists study the time complexity and feasibility of learning. In computational learning theory, a computation is considered feasible if it can be done in polynomial time. There are two kinds of time complexity results: Positive results show that a certain class of functions can be learned in polynomial time. Negative results show that certain classes cannot be learned in polynomial time.

Approaches

Machine learning approaches are traditionally divided into three broad categories, which correspond to learning paradigms, depending on the nature of the "signal" or "feedback" available to the learning system:

- Supervised learning: The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs.
- Unsupervised learning: No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself (discovering hidden patterns in data) or a means towards an end (feature learning).

- **Reinforcement learning:** A computer program interacts with a dynamic environment in which it must perform a certain goal (such as driving a vehicle or playing a game against an opponent). As it navigates its problem space, the program is provided rewards as feedback, which it tries to maximize, thus resulting in the program learning from experience.^[2]

Although each algorithm has advantages and limitations, no single algorithm works for all problems.^{[43][44][45]}

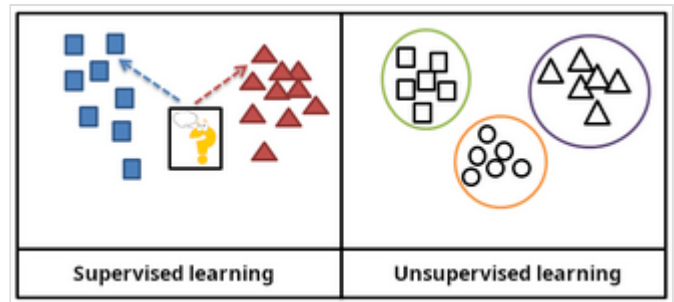
Supervised learning

Supervised learning algorithms build a mathematical model of a set of data that contains both the inputs and the desired outputs.^[46] The data, known as training data, consists of a set of training examples. Each training example has one or more inputs and the desired output, also known as a supervisory signal. In the mathematical model, each training example is represented by an array or vector, sometimes called a feature vector, and the training data is represented by a matrix. Through iterative optimisation of an objective function, supervised learning algorithms learn a function that can be used to predict the output associated with new inputs.^[47] An optimal function allows the algorithm to correctly determine the output for inputs that were not a part of the training data. An algorithm that improves the accuracy of its outputs or predictions over time is said to have learned to perform that task.^[15]

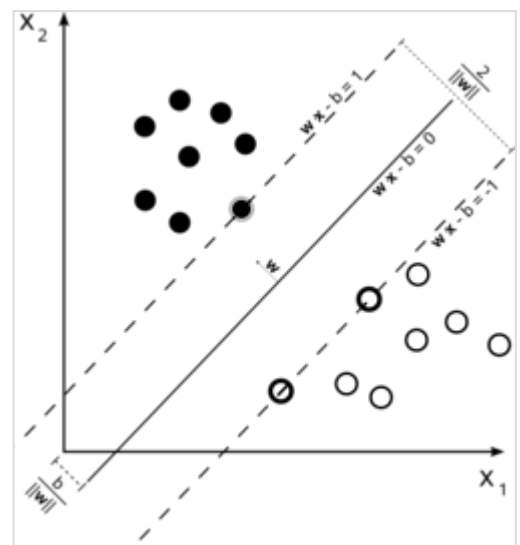
Types of supervised-learning algorithms include active learning, classification and regression.^[48] Classification

algorithms are used when the outputs are restricted to a limited set of values, while regression algorithms are used when the outputs can take any numerical value within a range. For example, in a classification algorithm that filters emails, the input is an incoming email, and the output is the folder in which to file the email. In contrast, regression is used for tasks such as predicting a person's height based on factors like age and genetics or forecasting future temperatures based on historical data.^[49]

Similarity learning is an area of supervised machine learning closely related to regression and classification, but the goal is to learn from examples using a similarity function that measures how similar or related two objects are. It has applications in ranking, recommendation systems, visual identity tracking, face verification, and speaker verification.



In supervised learning, the training data is labelled with the expected answers, while in unsupervised learning, the model identifies patterns or structures in unlabelled data.



A support-vector machine is a supervised learning model that divides the data into regions separated by a linear boundary. Here, the linear boundary divides the black circles from the white.

Unsupervised learning

Unsupervised learning algorithms find structures in data that has not been labelled, classified or categorised. Instead of responding to feedback, unsupervised learning algorithms identify commonalities in the data and react based on the presence or absence of such commonalities in each new piece of data. Central applications of unsupervised machine learning include clustering, dimensionality reduction,^[4] and density estimation.^[50]

Cluster analysis is the assignment of a set of observations into subsets (called *clusters*) so that observations within the same cluster are similar according to one or more predesignated criteria, while observations drawn from different clusters are dissimilar. Different clustering techniques make different assumptions on the structure of the data, often defined by some *similarity metric* and evaluated, for example, by *internal compactness*, or the similarity between members of the same cluster, and *separation*, the difference between clusters. Other methods are based on *estimated density* and *graph connectivity*.

A special type of unsupervised learning called self-supervised learning involves training a model by generating the supervisory signal from the data itself.^{[51][52]}

Dimensionality reduction

Dimensionality reduction is a process of reducing the number of random variables under consideration by obtaining a set of principal variables.^[53] In other words, it is a process of reducing the dimension of the feature set, also called the "number of features". Most of the dimensionality reduction techniques can be considered as either feature elimination or extraction. One of the popular methods of dimensionality reduction is principal component analysis (PCA). PCA involves changing higher-dimensional data (e.g., 3D) to a smaller space (e.g., 2D). The manifold hypothesis proposes that high-dimensional data sets lie along low-dimensional manifolds, and many dimensionality reduction techniques make this assumption, leading to the areas of manifold learning and manifold regularisation.

Semi-supervised learning

Semi-supervised learning falls between unsupervised learning (without any labelled training data) and supervised learning (with completely labelled training data). Some of the training examples are missing training labels, yet many machine-learning researchers have found that unlabelled data, when used in conjunction with a small amount of labelled data, can produce a considerable improvement in learning accuracy.

In weakly supervised learning, the training labels are noisy, limited, or imprecise; however, these labels are often cheaper to obtain, resulting in larger effective training sets.^[54]

Reinforcement learning

Reinforcement learning is an area of machine learning concerned with how software agents ought to take actions in an environment to maximise some notion of cumulative reward. Due to its generality, the field is studied in many other disciplines, such as game theory, control theory, operations research, information theory, simulation-based optimisation, multi-agent systems, swarm intelligence, statistics and genetic algorithms. In reinforcement learning, the environment is typically represented as a Markov decision

process (MDP). Many reinforcement learning algorithms use dynamic programming techniques.^[55] Reinforcement learning algorithms do not assume knowledge of an exact mathematical model of the MDP and are used when exact models are infeasible. Reinforcement learning algorithms are used in autonomous vehicles or in learning to play a game against a human opponent.

Other types

Other approaches have been developed which do not fit neatly into this three-fold categorisation, and sometimes more than one is used by the same machine learning system. For example, topic modelling, meta-learning.^[56]

Self-learning

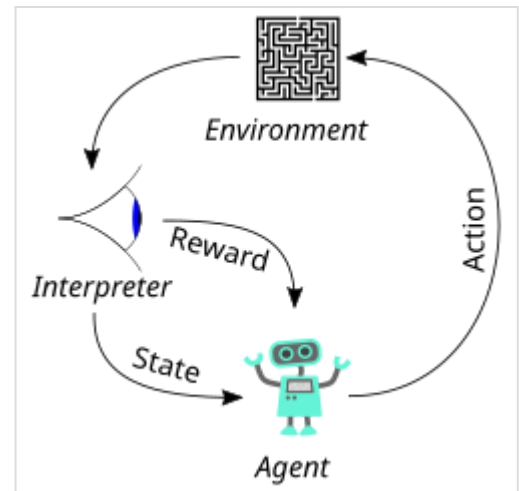
Self-learning, as a machine learning paradigm, was introduced in 1982 along with a neural network capable of self-learning, named *crossbar adaptive array* (CAA).^{[57][58]} It gives a solution to the problem learning without any external reward, by introducing emotion as an internal reward. Emotion is used as a state evaluation of a self-learning agent. The CAA self-learning algorithm computes, in a crossbar fashion, both decisions about actions and emotions (feelings) about consequence situations. The system is driven by the interaction between cognition and emotion.^[59] The self-learning algorithm updates a memory matrix $W = ||w(a,s)||$ such that in each iteration executes the following machine learning routine:

1. in situation s act a
2. receive a consequence situation s'
3. compute emotion of being in the consequence situation $v(s')$
4. update crossbar memory $w'(a,s) = w(a,s) + v(s')$

It is a system with only one input, situation, and only one output, action (or behaviour) a . There is neither a separate reinforcement input nor an advice input from the environment. The backpropagated value (secondary reinforcement) is the emotion toward the consequence situation. The CAA exists in two environments, one is the behavioural environment where it behaves, and the other is the genetic environment, wherefrom it initially and only once receives initial emotions about situations to be encountered in the behavioural environment. After receiving the genome (species) vector from the genetic environment, the CAA learns a goal-seeking behaviour in an environment that contains both desirable and undesirable situations.^[60]

Feature learning

Several learning algorithms aim at discovering better representations of the inputs provided during training.^[61] Classic examples include principal component analysis and cluster analysis. Feature learning algorithms, also called representation learning algorithms, often attempt to preserve the information in their input but also transform it in a way that makes it useful, often as a pre-processing step before performing classification or predictions. This technique allows reconstruction of the inputs coming from



In reinforcement learning, an agent takes actions in an environment: these produce a reward or a representation of the state, which is fed back to the agent.

the unknown data-generating distribution, while not being necessarily faithful to configurations that are implausible under that distribution. This replaces manual feature engineering, and allows a machine to both learn the features and use them to perform a specific task.

Feature learning can be either supervised or unsupervised. In supervised feature learning, features are learned using labelled input data. Examples include artificial neural networks, multilayer perceptrons, and supervised dictionary learning. In unsupervised feature learning, features are learned with unlabelled input data. Examples include dictionary learning, independent component analysis, autoencoders, matrix factorisation^[62] and various forms of clustering.^{[63][64][65]}

Manifold learning algorithms attempt to do so under the constraint that the learned representation is low-dimensional. Sparse coding algorithms attempt to do so under the constraint that the learned representation is sparse, meaning that the mathematical model has many zeros. Multilinear subspace learning algorithms aim to learn low-dimensional representations directly from tensor representations for multidimensional data, without reshaping them into higher-dimensional vectors.^[66] Deep learning algorithms discover multiple levels of representation, or a hierarchy of features, with higher-level, more abstract features defined in terms of (or generating) lower-level features. It has been argued that an intelligent machine learns a representation that disentangles the underlying factors of variation that explain the observed data.^[67]

Feature learning is motivated by the fact that machine learning tasks such as classification often require input that is mathematically and computationally convenient to process. However, real-world data such as images, video, and sensory data have not yielded attempts to algorithmically define specific features. An alternative is to discover such features or representations through examination, without relying on explicit algorithms.

Sparse dictionary learning

Sparse dictionary learning is a feature learning method where a training example is represented as a linear combination of basis functions and assumed to be a sparse matrix. The method is strongly NP-hard and difficult to solve approximately.^[68] A popular heuristic method for sparse dictionary learning is the k-SVD algorithm. Sparse dictionary learning has been applied in several contexts. In classification, the problem is to determine the class to which a previously unseen training example belongs. For a dictionary where each class has already been built, a new training example is associated with the class that is best sparsely represented by the corresponding dictionary. Sparse dictionary learning has also been applied in image denoising. The key idea is that a clean image patch can be sparsely represented by an image dictionary, but the noise cannot.^[69]

Anomaly detection

In data mining, anomaly detection, also known as outlier detection, is the identification of rare items, events or observations that raise suspicions by differing significantly from the majority of the data.^[70] Typically, the anomalous items represent an issue such as bank fraud, a structural defect, medical problems or errors in a text. Anomalies are referred to as outliers, novelties, noise, deviations and exceptions.^[71]

In particular, in the context of abuse and network intrusion detection, the interesting objects are often not rare, but unexpected bursts of inactivity. This pattern does not adhere to the common statistical definition of an outlier as a rare object. Many outlier detection methods (in particular, unsupervised algorithms) will fail on such data unless aggregated appropriately. Instead, a cluster analysis algorithm may be able to detect the micro-clusters formed by these patterns.^[72]

Three broad categories of anomaly detection techniques exist.^[73] Unsupervised anomaly detection techniques detect anomalies in an unlabelled test data set under the assumption that the majority of the instances in the data set are normal, by looking for instances that seem to fit the least to the remainder of the data set. Supervised anomaly detection techniques require a data set that has been labelled as "normal" and "abnormal" and involves training a classifier (the key difference from many other statistical classification problems is the inherently unbalanced nature of outlier detection). Semi-supervised anomaly detection techniques construct a model representing normal behaviour from a given normal training data set and then test the likelihood of a test instance being generated by the model.

Robot learning

Robot learning is inspired by a multitude of machine learning methods, starting from supervised learning, reinforcement learning,^{[74][75]} and finally meta-learning (e.g. MAML).

Association rules

Association rule learning is a rule-based machine learning method for discovering relationships between variables in large databases. It is intended to identify strong rules discovered in databases using some measure of "interestingness".^[76]

Rule-based machine learning is a general term for any machine learning method that identifies, learns, or evolves "rules" to store, manipulate or apply knowledge. The defining characteristic of a rule-based machine learning algorithm is the identification and utilisation of a set of relational rules that collectively represent the knowledge captured by the system. This is in contrast to other machine learning algorithms that commonly identify a singular model that can be universally applied to any instance in order to make a prediction.^[77] Rule-based machine learning approaches include learning classifier systems, association rule learning, and artificial immune systems.

Based on the concept of strong rules, Rakesh Agrawal, Tomasz Imieliński and Arun Swami introduced association rules for discovering regularities between products in large-scale transaction data recorded by point-of-sale (POS) systems in supermarkets.^[78] For example, the rule **{onions, potatoes} ⇒ {burger}** found in the sales data of a supermarket would indicate that if a customer buys onions and potatoes together, they are likely to also buy hamburger meat. Such information can be used as the basis for decisions about marketing activities such as promotional pricing or product placements. In addition to market basket analysis, association rules are employed today in application areas including Web usage mining, intrusion detection, continuous production, and bioinformatics. In contrast with sequence mining, association rule learning typically does not consider the order of items either within a transaction or across transactions.

Learning classifier systems (LCS) are a family of rule-based machine learning algorithms that combine a discovery component, typically a genetic algorithm, with a learning component, performing either supervised learning, reinforcement learning, or unsupervised learning. They seek to identify a set of

context-dependent rules that collectively store and apply knowledge in a piecewise manner to make predictions.^[79]

Inductive logic programming (ILP) is an approach to rule learning using logic programming as a uniform representation for input examples, background knowledge, and hypotheses. Given an encoding of the known background knowledge and a set of examples represented as a logical database of facts, an ILP system will derive a hypothesized logic program that entails all positive and no negative examples. Inductive programming is a related field that considers any kind of programming language for representing hypotheses (and not only logic programming), such as functional programs.

Inductive logic programming is particularly useful in bioinformatics and natural language processing. Gordon Plotkin and Ehud Shapiro laid the initial theoretical foundation for inductive machine learning in a logical setting.^{[80][81][82]} Shapiro built their first implementation (Model Inference System) in 1981: a Prolog program that inductively inferred logic programs from positive and negative examples.^[83] The term *inductive* here refers to philosophical induction, suggesting a theory to explain observed facts, rather than mathematical induction, proving a property for all members of a well-ordered set.

Models

A **machine learning model** is a type of mathematical model that, once "trained" on a given dataset, can be used to make predictions or classifications on new data. During training, a learning algorithm iteratively adjusts the model's internal parameters to minimise errors in its predictions.^[84] By extension, the term "model" can refer to several levels of specificity, from a general class of models and their associated learning algorithms to a fully trained model with all its internal parameters tuned.^[85]

Various types of models have been used and researched for machine learning systems, picking the best model for a task is called model selection.

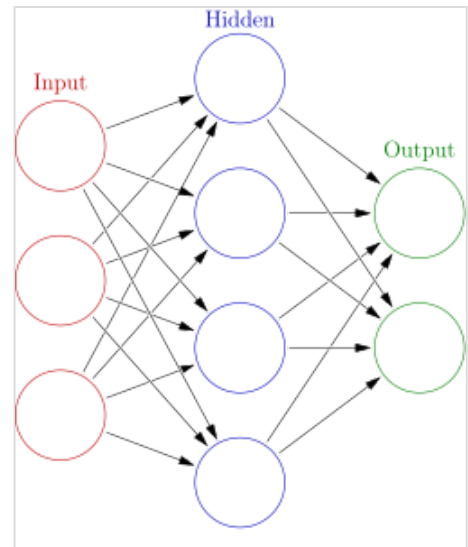
Artificial neural networks

Artificial neural networks (ANNs), or connectionist systems, are computing systems vaguely inspired by the biological neural networks that constitute animal brains. Such systems "learn" to perform tasks by considering examples, generally without being programmed with any task-specific rules.

An ANN is a model based on a collection of connected units or nodes called "artificial neurons", which loosely model the neurons in a biological brain. Each connection, like the synapses in a biological brain, can transmit information, a "signal", from one artificial neuron to another. An artificial neuron that receives a signal can process it and then signal additional artificial neurons connected to it. In common ANN implementations, the signal at a connection between artificial neurons is a real number, and the output of each artificial neuron is computed by some non-linear function of the sum of its inputs. The connections between artificial neurons are called "edges". Artificial neurons and edges typically have a weight that adjusts as learning proceeds. The weight increases or decreases the strength of the signal at a connection. Artificial neurons may have a threshold such that the signal is only sent if the aggregate signal crosses that threshold. Typically, artificial neurons are aggregated into layers. Different layers may perform different kinds of transformations on their inputs. Signals travel from the first layer (the input layer) to the last layer (the output layer), possibly after traversing the layers multiple times.

The original goal of the ANN approach was to solve problems in the same way that a human brain would. However, over time, attention moved to performing specific tasks, leading to deviations from biology. Artificial neural networks have been used on a variety of tasks, including computer vision, speech recognition, machine translation, social network filtering, playing board and video games and medical diagnosis.

Deep learning consists of multiple hidden layers in an artificial neural network. This approach tries to model the way the human brain processes light and sound into vision and hearing. Some successful applications of deep learning are computer vision and speech recognition.^[86]



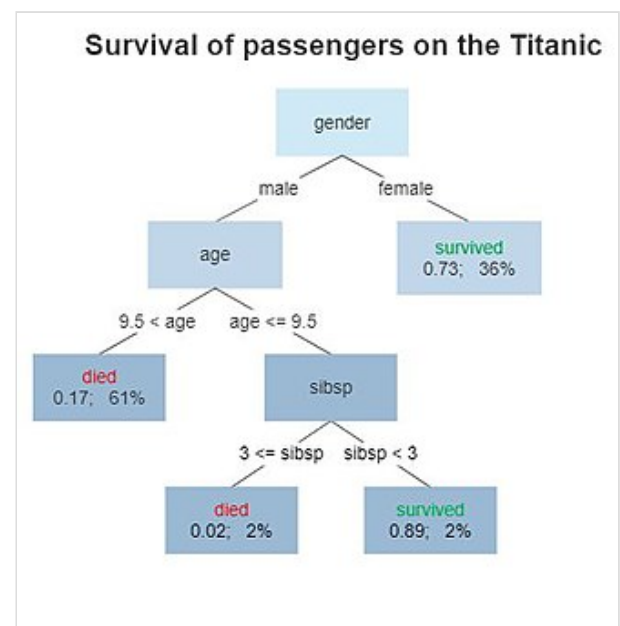
An artificial neural network is an interconnected group of nodes, akin to the vast network of neurons in a brain. Here, each circular node represents an artificial neuron and an arrow represents a connection from the output of one artificial neuron to the input of another.

Decision trees

Decision tree learning uses a decision tree as a predictive model to go from observations about an item (represented in the branches) to conclusions about the item's target value (represented in the leaves). It is one of the predictive modelling approaches used in statistics, data mining, and machine learning. Tree models where the target variable can take a discrete set of values are called classification trees; in these tree structures, leaves represent class labels, and branches represent conjunctions of features that lead to those class labels. Decision trees where the target variable can take continuous values (typically real numbers) are called regression trees. In decision analysis, a decision tree can be used to visually and explicitly represent decisions and decision making. In data mining, a decision tree describes data, but the resulting classification tree can be an input for decision-making.

Random forest regression

Random forest regression (RFR) falls under the umbrella of decision tree-based models. RFR is an ensemble learning method that builds multiple decision trees and averages their predictions to improve accuracy and to avoid overfitting. To build decision trees, RFR uses bootstrapped sampling; for instance, each decision tree is trained on random data from the training set. This random selection of RFR for training enables the model to reduce biased predictions and achieve a higher degree of accuracy. RFR generates independent decision trees, and it can work on single-output data as well as multiple regressor tasks. This makes RFR compatible to be use in various applications.^{[87][88]}



A decision tree showing survival probability of passengers on the Titanic

Support-vector machines

Support-vector machines (SVMs), also known as support-vector networks, are a set of related supervised learning methods used for classification and regression. Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that predicts whether a new example falls into one category.^[89] An SVM training algorithm is a non-probabilistic, binary, linear classifier, although methods such as Platt scaling exist to use SVM in a probabilistic classification setting. In addition to performing linear classification, SVMs can efficiently perform a non-linear classification using what is called the kernel trick, implicitly mapping their inputs into high-dimensional feature spaces.

Regression analysis

Regression analysis encompasses a large variety of statistical methods to estimate the relationship between input variables and their associated features. Its most common form is linear regression, where a single line is drawn to best fit the given data according to a mathematical criterion such as ordinary least squares. The latter is often extended by regularisation methods to mitigate overfitting and bias, as in ridge regression. When dealing with non-linear problems, go-to models include polynomial regression (for example, used for trendline fitting in Microsoft Excel^[90]), logistic regression (often

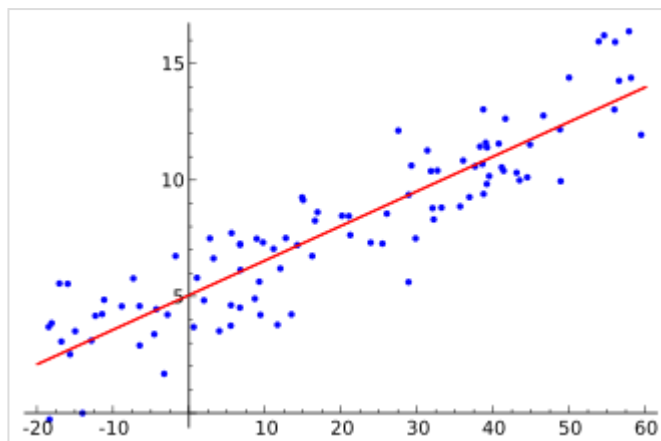


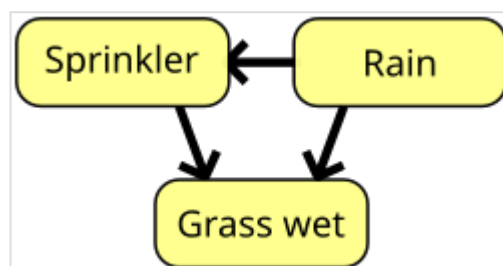
Illustration of linear regression on a data set

used in statistical classification) or even kernel regression, which introduces non-linearity by taking advantage of the kernel trick to implicitly map input variables to higher-dimensional space.

Multivariate linear regression extends the concept of linear regression to handle multiple dependent variables simultaneously. This approach estimates the relationships between a set of input variables and several output variables by fitting a multidimensional linear model. It is particularly useful in scenarios where outputs are interdependent or share underlying patterns, such as predicting multiple economic indicators or reconstructing images,^[91] which are inherently multi-dimensional.

Bayesian networks

A Bayesian network, belief network, or directed acyclic graphical model is a probabilistic graphical model that represents a set of random variables and their conditional independence with a directed acyclic graph (DAG). For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms exist that perform inference and learning. Bayesian networks that model sequences of variables, like



A simple Bayesian network. Rain influences whether the sprinkler is activated, and both rain and the sprinkler influence whether the grass is wet.

speech signals or protein sequences, are called dynamic Bayesian networks. Generalisations of Bayesian networks that can represent and solve decision problems under uncertainty are called influence diagrams.

Gaussian processes

A Gaussian process is a stochastic process in which every finite collection of the random variables in the process has a multivariate normal distribution, and it relies on a pre-defined covariance function, or kernel, that models how pairs of points relate to each other depending on their locations.

Given a set of observed points, or input–output examples, the distribution of the (unobserved) output of a new point as a function of its input data can be directly computed by looking at the observed points and the covariances between those points and the new, unobserved point.

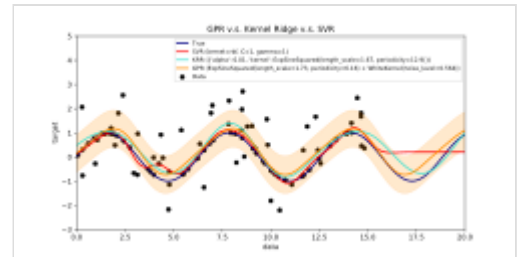
Gaussian processes are popular surrogate models in Bayesian optimisation used to do hyperparameter optimisation.

Genetic algorithms

A genetic algorithm (GA) is a search algorithm and heuristic technique that mimics the process of natural selection, using methods such as mutation and crossover to generate new genotypes in the hope of finding good solutions to a given problem. In machine learning, genetic algorithms were used in the 1980s and 1990s.^{[93][94]} Conversely, machine learning techniques have been used to improve the performance of genetic and evolutionary algorithms.^[95]

Belief functions

The theory of belief functions, also referred to as evidence theory or Dempster–Shafer theory, is a general framework for reasoning with uncertainty, with understood connections to other frameworks such as probability, possibility and imprecise probability theories. These theoretical frameworks can be thought of as a kind of learner and have some analogous properties of how evidence is combined (e.g., Dempster's rule of combination), just like how in a pmf-based Bayesian approach would combine probabilities.^[96] However, there are many caveats to these beliefs functions when compared to Bayesian approaches to incorporate ignorance and uncertainty quantification. These belief function approaches that are implemented within the machine learning domain typically leverage a fusion approach of various ensemble methods to better handle the learner's decision boundary, low samples, and ambiguous class issues that standard machine learning approach tend to have difficulty resolving.^{[97][6]} However, the computational complexity of these algorithms is dependent on the number of propositions (classes), and can lead to a much higher computation time when compared to other machine learning approaches.



An example of Gaussian Process Regression (prediction) compared with other regression models^[92]

Rule-based models

Rule-based machine learning (RBML) is a branch of machine learning that automatically discovers and learns 'rules' from data. It provides interpretable models, making it useful for decision-making in fields like healthcare, fraud detection, and cybersecurity. Key RBML techniques includes learning classifier systems,^[98] association rule learning,^[99] artificial immune systems,^[100] and other similar models. These methods extract patterns from data and evolve rules over time.

Training models

Typically, machine learning models require a high quantity of reliable data to perform accurate predictions. When training a machine learning model, machine learning engineers need to target and collect a large and representative sample of data. Data from the training set can be as varied as a corpus of text, a collection of images, sensor data, and data collected from individual users of a service. Overfitting is something to watch out for when training a machine learning model. Trained models derived from biased or non-evaluated data can result in skewed or undesired predictions. Biased models may result in detrimental outcomes, thereby furthering the negative impacts on society or objectives. Algorithmic bias is a potential result of data not being fully prepared for training. Machine learning ethics is becoming a field of study and, notably, becoming integrated within machine learning engineering teams.

Federated learning

Federated learning is an adapted form of distributed artificial intelligence to train machine learning models that decentralises the training process, allowing for users' privacy to be maintained by not needing to send their data to a centralised server. This also increases efficiency by decentralising the training process to many devices. For example, Gboard uses federated machine learning to train search query prediction models on users' mobile phones without having to send individual searches back to Google.^[101]

Applications

There are many applications for machine learning, including:

- Agriculture
- Anatomy
- Adaptive website
- Affective computing
- Astronomy
- Automated decision-making
- Banking
- Behaviorism
- Bioinformatics
- Brain–machine interfaces
- Cheminformatics
- Citizen Science
- Climate Science

- Computer networks
- Computer vision
- Credit-card fraud detection
- Data quality
- DNA sequence classification
- Economics
- Financial data analysis^[102]
- General game playing
- Handwriting recognition
- Healthcare
- Information retrieval
- Insurance
- Internet fraud detection
- Investment management^[103]
- Knowledge graph embedding
- Linguistics
- Machine learning control
- Machine perception
- Machine translation
- Material Engineering
- Marketing
- Medical diagnosis
- Natural language processing
- Natural language understanding
- Online advertising
- Optimisation
- Recommender systems
- Robot locomotion
- Search engines
- Sentiment analysis
- Sequence mining
- Software engineering
- Speech recognition
- Structural health monitoring
- Syntactic pattern recognition
- Telecommunications
- Theorem proving
- Time-series forecasting
- Tomographic reconstruction^[104]
- User behaviour analytics

In 2006, the media-services provider Netflix held the first "Netflix Prize" competition to find a program to better predict user preferences and improve the accuracy of its existing Cinematch movie recommendation algorithm by at least 10%. A joint team made up of researchers from AT&T Labs-Research in collaboration with the teams Big Chaos and Pragmatic Theory built an ensemble model to win the Grand Prize in 2009 for \$1 million.^[105] Shortly after the prize was awarded, Netflix realised that

viewers' ratings were not the best indicators of their viewing patterns ("everything is a recommendation") and they changed their recommendation engine accordingly.^[106] In 2010, an article in *The Wall Street Journal* noted the use of machine learning by Rebellion Research to predict the 2008 financial crisis.^[107] In 2012, co-founder of Sun Microsystems, Vinod Khosla, predicted that 80% of medical doctors jobs would be lost in the next two decades to automated machine learning medical diagnostic software.^[108] In 2014, it was reported that a machine learning algorithm had been applied in the field of art history to study fine art paintings and that it may have revealed previously unrecognised influences among artists.^[109] In 2019 Springer Nature published the first research book created using machine learning.^[110] In 2020, machine learning technology was used to help make diagnoses and aid researchers in developing a cure for COVID-19.^[111] Machine learning was recently applied to predict the pro-environmental behaviour of travellers.^[112] Recently, machine learning technology was also applied to optimise smartphone's performance and thermal behaviour based on the user's interaction with the phone.^{[113][114][115]} When applied correctly, machine learning algorithms (MLAs) can utilise a wide range of company characteristics to predict stock returns without overfitting. By employing effective feature engineering and combining forecasts, MLAs can generate results that far surpass those obtained from basic linear techniques like OLS.^[116]

Recent advancements in machine learning have extended into the field of quantum chemistry, where novel algorithms now enable the prediction of solvent effects on chemical reactions, thereby offering new tools for chemists to tailor experimental conditions for optimal outcomes.^[117]

Machine Learning is becoming a useful tool to investigate and predict evacuation decision-making in large-scale and small-scale disasters. Different solutions have been tested to predict if and when householders decide to evacuate during wildfires and hurricanes.^{[118][119][120]} Other applications have been focusing on pre evacuation decisions in building fires.^{[121][122]}

Limitations

Although machine learning has been transformative in some fields, machine-learning programs often fail to deliver expected results.^{[123][124][125]} Reasons for this are numerous: lack of (suitable) data, lack of access to the data, data bias, privacy problems, badly chosen tasks and algorithms, wrong tools and people, lack of resources, and evaluation problems.^[126]

The "black box theory" poses another yet significant challenge. Black box refers to a situation where the algorithm or the process of producing an output is entirely opaque, meaning that even the coders of the algorithm cannot audit the pattern that the machine extracted from the data.^[127] The House of Lords Select Committee, which claimed that such an "intelligence system" that could have a "substantial impact on an individual's life" would not be considered acceptable unless it provided "a full and satisfactory explanation for the decisions" it makes.^[127]

In 2018, a self-driving car from Uber failed to detect a pedestrian, who was killed after a collision.^[128] Attempts to use machine learning in healthcare with the IBM Watson system failed to deliver even after years of time and billions of dollars invested.^{[129][130]} Microsoft's Bing Chat chatbot has been reported to produce hostile and offensive response against its users.^[131]

Machine learning has been used as a strategy to update the evidence related to a systematic review and increased reviewer burden related to the growth of biomedical literature. While it has improved with training sets, it has not yet developed sufficiently to reduce the workload burden without limiting the necessary sensitivity for the findings research itself.^[132]

Explainability

Explainable AI (XAI), or Interpretable AI, or Explainable Machine Learning (XML), is artificial intelligence (AI) in which humans can understand the decisions or predictions made by the AI.^[133] It contrasts with the "black box" concept in machine learning where even its designers cannot explain why an AI arrived at a specific decision.^[134] By refining the mental models of users of AI-powered systems and dismantling their misconceptions, XAI promises to help users perform more effectively. XAI may be an implementation of the social right to explanation.

Overfitting

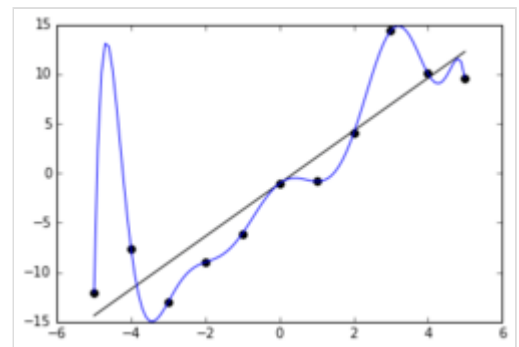
Settling on a bad, overly complex theory gerrymandered to fit all the past training data is known as overfitting. Many systems attempt to reduce overfitting by rewarding a theory in accordance with how well it fits the data but penalising the theory in accordance with how complex the theory is.^[135]

Other limitations and vulnerabilities

Learners can also be disappointed by "learning the wrong lesson". A toy example is that an image classifier trained only on pictures of brown horses and black cats might conclude that all brown patches are likely to be horses.^[136] A real-world example is that, unlike humans, current image classifiers often do not primarily make judgments from the spatial relationship between components of the picture, and they learn relationships between pixels that humans are oblivious to, but that still correlate with images of certain types of real objects. Modifying these patterns on a legitimate image can result in "adversarial" images that the system misclassifies.^{[137][138]}

Adversarial vulnerabilities can also result in nonlinear systems or from non-pattern perturbations. For some systems, it is possible to change the output by only changing a single adversarially chosen pixel.^[139] Machine learning models are often vulnerable to manipulation or evasion via adversarial machine learning.^[140]

Researchers have demonstrated how backdoors can be placed undetectably into classifying (e.g., for categories "spam" and "not spam" of posts) machine learning models that are often developed or trained by third parties. Parties can change the classification of any input, including in cases for which a type of data/software transparency is provided, possibly including white-box access.^{[141][142][143]}



The blue line could be an example of overfitting a linear function due to random noise.

Model assessments

Classification of machine learning models can be validated by accuracy estimation techniques like the holdout method, which splits the data into a training and test set (conventionally 2/3 training set and 1/3 test set designation) and evaluates the performance of the training model on the test set. In comparison, the K-fold-cross-validation method randomly partitions the data into K subsets and then K experiments are performed each considering 1 subset for evaluation and the remaining K-1 subsets for training the model. In addition to the holdout and cross-validation methods, bootstrap, which samples n instances with replacement from the dataset, can be used to assess model accuracy.^[144]

In addition to overall accuracy, investigators frequently report sensitivity and specificity, meaning true positive rate (TPR) and true negative rate (TNR), respectively. Similarly, investigators sometimes report the false positive rate (FPR) as well as the false negative rate (FNR). However, these rates are ratios that fail to reveal their numerators and denominators. Receiver operating characteristic (ROC), along with the accompanying Area Under the ROC Curve (AUC), offer additional tools for classification model assessment. Higher AUC is associated with a better performing model.^[145]

Ethics

The ethics of artificial intelligence covers a broad range of topics within AI that are considered to have particular ethical stakes.^[146] This includes algorithmic biases, fairness, accountability, transparency, privacy, and regulation, particularly where systems influence or automate human decision-making. It also covers various emerging or potential future challenges such as machine ethics (how to make machines that behave ethically), lethal autonomous weapon systems, arms race dynamics, AI safety and alignment, technological unemployment, AI-enabled misinformation,^[147] how to treat certain AI systems if they have a moral status (AI welfare and rights), artificial superintelligence and existential risks.^[146]

Some application areas may also have particularly important ethical implications, like healthcare, education, criminal justice, or the military.

Bias

Different machine learning approaches can suffer from different data biases. A machine learning system trained specifically on current customers may not be able to predict the needs of new customer groups that are not represented in the training data. When trained on human-made data, machine learning is likely to pick up the constitutional and unconscious biases already present in society.^[148]

Systems that are trained on datasets collected with biases may exhibit these biases upon use (algorithmic bias), thus digitising cultural prejudices.^[149] For example, in 1988, the UK's Commission for Racial Equality found that St. George's Medical School had been using a computer program trained from data of previous admissions staff and this program had denied nearly 60 candidates who were found to either be women or have non-European-sounding names.^[148] Using job hiring data from a firm with racist hiring policies may lead to a machine learning system duplicating the bias by scoring job applicants by

similarity to previous successful applicants.^{[150][151]} Another example includes predictive policing company Geolitica's predictive algorithm that resulted in "disproportionately high levels of over-policing in low-income and minority communities" after being trained with historical crime data.^[152]

While responsible collection of data and documentation of algorithmic rules used by a system is considered a critical part of machine learning, some researchers blame the lack of participation and representation of minority populations in the field of AI for machine learning's vulnerability to biases.^[153] In fact, according to research carried out by the Computing Research Association in 2021, "female faculty make up just 16.1%" of all faculty members who focus on AI among several universities around the world.^[154] Furthermore, among the group of "new U.S. resident AI PhD graduates," 45% identified as white, 22.4% as Asian, 3.2% as Hispanic, and 2.4% as African American, which further demonstrates a lack of diversity in the field of AI.^[154]

Language models learned from data have been shown to contain human-like biases.^{[155][156]} Because human languages contain biases, machines trained on language *corpora* will necessarily also learn these biases.^{[157][158]} In 2016, Microsoft tested Tay, a chatbot that learned from Twitter, and it quickly picked up racist and sexist language.^[159]

In an experiment carried out by ProPublica, an investigative journalism organisation, a machine learning algorithm's insight into the recidivism rates among prisoners falsely flagged "black defendants high risk twice as often as white defendants".^[152] In 2015, Google Photos once tagged a couple of black people as gorillas, which caused controversy. The gorilla label was subsequently removed, and in 2023, it still cannot recognise gorillas.^[160] Similar issues with recognising non-white people have been found in many other systems.^[161]

Because of such challenges, the effective use of machine learning may take longer to be adopted in other domains.^[162] Concern for fairness in machine learning, that is, reducing bias in machine learning and propelling its use for human good, is increasingly expressed by artificial intelligence scientists, including Fei-Fei Li, who said that "[t]here's nothing artificial about AI. It's inspired by people, it's created by people, and—most importantly—it impacts people. It is a powerful tool we are only just beginning to understand, and that is a profound responsibility."^[163]

Financial incentives

There are concerns among health care professionals that these systems might not be designed in the public's interest but as income-generating machines. This is especially true in the United States, where there is a long-standing ethical dilemma of improving health care, but also increasing profits. For example, the algorithms could be designed to provide patients with unnecessary tests or medication in which the algorithm's proprietary owners hold stakes. There is potential for machine learning in health care to provide professionals with an additional tool to diagnose, medicate, and plan recovery paths for patients, but this requires these biases to be mitigated.^[164]

Hardware

Since the 2010s, advances in both machine learning algorithms and computer hardware have led to more efficient methods for training deep neural networks (a particular narrow subdomain of machine learning) that contain many layers of nonlinear hidden units.^[165] By 2019, graphics processing units (GPUs), often

with AI-specific enhancements, had displaced CPUs as the dominant method of training large-scale commercial cloud AI.^[166] OpenAI estimated the hardware compute used in the largest deep learning projects from AlexNet (2012) to AlphaZero (2017), and found a 300,000-fold increase in the amount of compute required, with a doubling-time trendline of 3.4 months.^{[167][168]}

Tensor Processing Units (TPUs)

Tensor Processing Units (TPUs) are specialised hardware accelerators developed by Google specifically for machine learning workloads. Unlike general-purpose GPUs and FPGAs, TPUs are optimised for tensor computations, making them particularly efficient for deep learning tasks such as training and inference. They are widely used in Google Cloud AI services and large-scale machine learning models like Google's DeepMind AlphaFold and large language models. TPUs leverage matrix multiplication units and high-bandwidth memory to accelerate computations while maintaining energy efficiency.^[169] Since their introduction in 2016, TPUs have become a key component of AI infrastructure, especially in cloud-based environments.

Neuromorphic computing

Neuromorphic computing refers to a class of computing systems designed to emulate the structure and functionality of biological neural networks. These systems may be implemented through software-based simulations on conventional hardware or through specialised hardware architectures.^[170]

Physical neural networks

A physical neural network is a specific type of neuromorphic hardware that relies on electrically adjustable materials, such as memristors, to emulate the function of neural synapses. The term "physical neural network" highlights the use of physical hardware for computation, as opposed to software-based implementations. It broadly refers to artificial neural networks that use materials with adjustable resistance to replicate neural synapses.^{[171][172]}

Embedded machine learning

Embedded machine learning is a sub-field of machine learning where models are deployed on embedded systems with limited computing resources, such as wearable computers, edge devices and microcontrollers.^{[173][174][175][176]} Running models directly on these devices eliminates the need to transfer and store data on cloud servers for further processing, thereby reducing the risk of data breaches, privacy leaks and theft of intellectual property, personal data and business secrets. Embedded machine learning can be achieved through various techniques, such as hardware acceleration,^{[177][178]} approximate computing,^[179] and model optimisation.^{[180][181]} Common optimisation techniques include pruning, quantisation, knowledge distillation, low-rank factorisation, network architecture search, and parameter sharing.

Artificial intelligence

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.^[1]

High-profile applications of AI include advanced web search engines, chatbots, virtual assistants, autonomous vehicles, and play and analysis in strategy games (e.g., chess and Go). Since the 2020s, generative AI has become widely available to generate images, audio, and videos from text prompts.

The traditional goals of AI research include learning, reasoning, knowledge representation, planning, natural language processing, and perception, as well as support for robotics.^[a] To reach these goals, AI researchers have used techniques including state space search and mathematical optimization, formal logic, artificial neural networks, and methods based on statistics, operations research, and economics.^[b] AI also draws upon psychology, linguistics, philosophy, neuroscience, and other fields.^[2] Some companies, such as OpenAI, Google DeepMind and Meta, aim to create artificial general intelligence (AGI) – AI that can complete virtually any cognitive task at least as well as a human.^[3]

Artificial intelligence was founded as an academic discipline in 1956,^[4] and the field went through multiple cycles of optimism throughout its history,^{[5][6]} followed by periods of disappointment and loss of funding, known as AI winters.^{[7][8]} Funding and interest increased substantially after 2012, when graphics processing units began being used to accelerate neural networks, and deep learning outperformed previous AI techniques.^[9] This growth accelerated further after 2017 with the transformer architecture.^[10] In the 2020s, an AI boom has coincided with advances in generative AI, which allowed for the creation and modification of media. In addition to AI safety and unintended consequences and harms from the use of AI, ethical concerns, AI's long-term effects, and potential existential risks have prompted discussions of AI regulation.

Goals

The general problem of simulating (or creating) intelligence has been broken into subproblems. These consist of particular traits or capabilities that researchers expect an intelligent system to display. The traits described below have received the most attention and cover the scope of AI research.^[a]

Reasoning and problem-solving

Early researchers developed algorithms that imitated step-by-step reasoning that humans use when they solve puzzles or make logical deductions.^[11] By the late 1980s and 1990s, methods were developed for dealing with uncertain or incomplete information, employing concepts from probability and economics.^[12]

Many of these algorithms are insufficient for solving large reasoning problems because they experience a "combinatorial explosion": They become exponentially slower as the problems grow.^[13] Even humans rarely use the step-by-step deduction that early AI research could model. They solve most of their problems using fast, intuitive judgments.^[14] Accurate and efficient reasoning is an unsolved problem.

Knowledge representation

Knowledge representation and knowledge engineering^[15] allow AI programs to answer questions intelligently and make deductions about real-world facts. Formal knowledge representations are used in content-based indexing and retrieval,^[16] scene interpretation,^[17] clinical decision support,^[18] knowledge discovery (mining "interesting" and actionable inferences from large databases),^[19] and other areas.^[20]

A knowledge base is a body of knowledge represented in a form that can be used by a program. An ontology is the set of objects, relations, concepts, and properties used by a particular domain of knowledge.^[21] Knowledge bases need to represent things such as objects, properties, categories, and relations between objects;^[22] situations, events, states, and time;^[23] causes and effects;^[24] knowledge about knowledge (what we know about what other people know);^[25] default reasoning (things that humans assume are true until they are told differently and will remain true even when other facts are changing);^[26] and many other aspects and domains of knowledge.

Among the most difficult problems in knowledge representation are the breadth of commonsense knowledge (the set of atomic facts that the average person knows is enormous);^[27] and the sub-symbolic form of most commonsense knowledge (much of what people know is not represented as "facts" or "statements" that they could express verbally).^[14] There is also the difficulty of knowledge acquisition, the problem of obtaining knowledge for AI applications.^[c]

Planning and decision-making

An "agent" is any entity (artificial or not) that perceives and takes actions in the world. A rational agent has goals or preferences and takes actions to make them happen.^{[d][30]} In automated planning, the agent has a specific goal.^[31] In automated decision-making, the agent has preferences—there are some situations it would prefer to be in, and some situations it is trying to avoid. The decision-making agent assigns a number to each situation (called the "utility") that measures how much the agent prefers it. For each possible action, it can calculate the "expected utility": the utility of all possible outcomes of the action, weighted by the probability that the outcome will occur. It can then choose the action with the maximum expected utility.^[32]



An ontology represents knowledge as a set of concepts within a domain and the relationships between those concepts.

In classical planning, the agent knows exactly what the effect of any action will be.^[33] In most real-world problems, however, the agent may not be certain about the situation they are in (it is "unknown" or "unobservable") and it may not know for certain what will happen after each possible action (it is not "deterministic"). It must choose an action by making a probabilistic guess and then reassess the situation to see if the action worked.^[34]

Alongside thorough testing and improvement based on previous decisions, having an explanation for why the agent took certain decisions is a way to build trust, especially when the decisions have to be relied upon.^[35]

In some problems, the agent's preferences may be uncertain, especially if there are other agents or humans involved. These can be learned (e.g., with inverse reinforcement learning), or the agent can seek information to improve its preferences.^[36] Information value theory can be used to weigh the value of exploratory or experimental actions.^[37] The space of possible future actions and situations is typically intractably large, so the agents must take actions and evaluate situations while being uncertain of what the outcome will be.

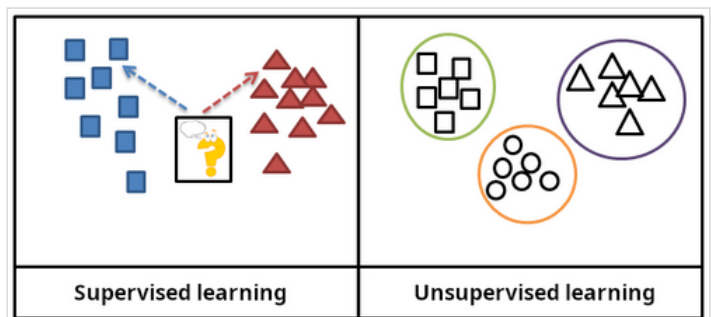
A Markov decision process has a transition model that describes the probability that a particular action will change the state in a particular way and a reward function that supplies the utility of each state and the cost of each action. A policy associates a decision with each possible state. The policy could be calculated (e.g., by iteration), be heuristic, or it can be learned.^[38]

Game theory describes the rational behavior of multiple interacting agents and is used in AI programs that make decisions that involve other agents.^[39]

Learning

Machine learning is the study of programs that can improve their performance on a given task automatically.^[40] It has been a part of AI from the beginning.^[e]

There are several kinds of machine learning. Unsupervised learning analyzes a stream of data and finds patterns and makes predictions without any other guidance.^[43] Supervised learning requires labeling the training data with the expected answers, and comes in two main varieties: classification (where the program must learn to predict what category the input belongs in) and regression (where the program must deduce a numeric function based on numeric input).^[44]



In supervised learning, the training data is labelled with the expected answers, while in unsupervised learning, the model identifies patterns or structures in unlabelled data.

In reinforcement learning, the agent is rewarded for good responses and punished for bad ones. The agent learns to choose responses that are classified as "good".^[45] Transfer learning is when the knowledge gained from one problem is applied to a new problem.^[46] Deep learning is a type of machine learning that runs inputs through biologically inspired artificial neural networks for all of these types of learning.^[47]

Computational learning theory can assess learners by computational complexity, by sample complexity (how much data is required), or by other notions of optimization.^[48]

Natural language processing

Natural language processing (NLP) allows programs to read, write and communicate in human languages.^[49] Specific problems include speech recognition, speech synthesis, machine translation, information extraction, information retrieval and question answering.^[50]

Early work, based on Noam Chomsky's generative grammar and semantic networks, had difficulty with word-sense disambiguation^[f] unless restricted to small domains called "micro-worlds" (due to the common sense knowledge problem^[27]). Margaret Masterman believed that it was meaning and not grammar that was the key to understanding languages, and that thesauri and not dictionaries should be the basis of computational language structure.

Modern deep learning techniques for NLP include word embedding (representing words, typically as vectors encoding their meaning),^[51] transformers (a deep learning architecture using an attention mechanism),^[52] and others.^[53] In 2019, generative pre-trained transformer (or "GPT") language models began to generate coherent text,^{[54][55]} and by 2023, these models were able to get human-level scores on the bar exam, SAT test, GRE test, and many other real-world applications.^[56]

Perception

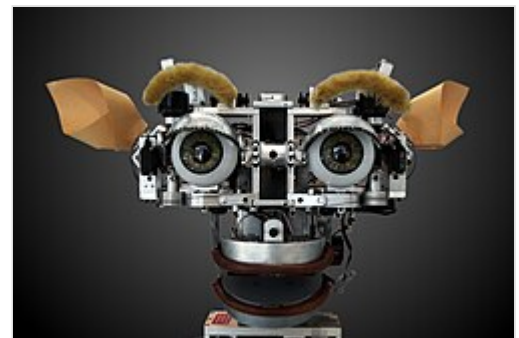
Machine perception is the ability to use input from sensors (such as cameras, microphones, wireless signals, active lidar, sonar, radar, and tactile sensors) to deduce aspects of the world. Computer vision is the ability to analyze visual input.^[57]

The field includes speech recognition,^[58] image classification,^[59] facial recognition, object recognition,^[60] object tracking,^[61] and robotic perception.^[62]

Social intelligence

Affective computing is a field that comprises systems that recognize, interpret, process, or simulate human feeling, emotion, and mood.^[64] For example, some virtual assistants are programmed to speak conversationally or even to banter humorously; it makes them appear more sensitive to the emotional dynamics of human interaction, or to otherwise facilitate human-computer interaction.

However, this tends to give naïve users an unrealistic conception of the intelligence of existing computer agents.^[65] Moderate successes related to affective computing include textual sentiment analysis and, more recently, multimodal sentiment analysis, wherein AI classifies the effects displayed by a videotaped subject.^[66]



Kismet, a robot head made in the 1990s, is a machine that can recognize and simulate emotions.^[63]

General intelligence

A machine with artificial general intelligence would be able to solve a wide variety of problems with breadth and versatility similar to human intelligence.^[67]

Techniques

AI research uses a wide variety of techniques to accomplish the goals above.^[b]

Search and optimization

There are two different kinds of search used in AI: state space search and local search:

State space search

State space search searches through a tree of possible states to try to find a goal state.^[68] For example, planning algorithms search through trees of goals and subgoals, attempting to find a path to a target goal, a process called means-ends analysis.^[69]

Simple exhaustive searches^[70] are rarely sufficient for most real-world problems: the search space (the number of places to search) quickly grows to astronomical numbers. The result is a search that is too slow or never completes.^[13] "Heuristics" or "rules of thumb" can help prioritize choices that are more likely to reach a goal.^[71]

Adversarial search is used for game-playing programs, such as chess or Go. It searches through a tree of possible moves and countermoves, looking for a winning position.^[72]

Local search

Local search uses mathematical optimization to find a solution to a problem. It begins with some form of guess and refines it incrementally.^[73]

Gradient descent is a type of local search that optimizes a set of numerical parameters by incrementally adjusting them to minimize a loss function. Variants of gradient descent are commonly used to train neural networks,^[74] through the backpropagation algorithm.

Another type of local search is evolutionary computation, which aims to iteratively improve a set of candidate solutions by "mutating" and "recombining" them, selecting only the fittest to survive each generation.^[75]

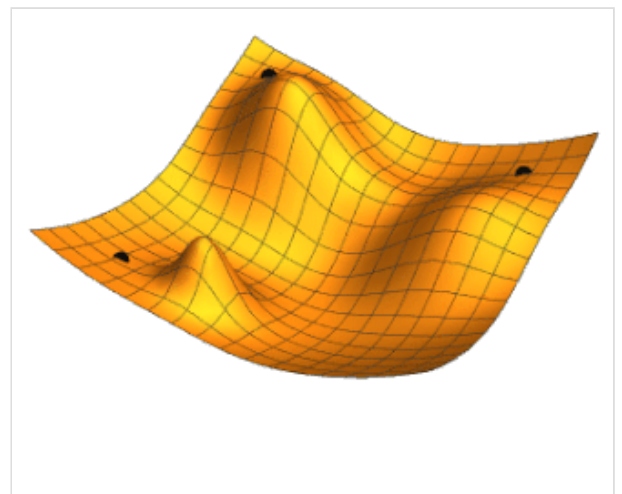


Illustration of gradient descent for three different starting points; two parameters (represented by the plan coordinates) are adjusted in order to minimize the loss function (the height).

Distributed search processes can coordinate via swarm intelligence algorithms. Two popular swarm algorithms used in search are particle swarm optimization (inspired by bird flocking) and ant colony optimization (inspired by ant trails).^[76]

Logic

Formal logic is used for reasoning and knowledge representation.^[77] Formal logic comes in two main forms: propositional logic (which operates on statements that are true or false and uses logical connectives such as "and", "or", "not" and "implies")^[78] and predicate logic (which also operates on objects, predicates and relations and uses quantifiers such as "*Every X is a Y*" and "*There are some Xs that are Ys*").^[79]

Deductive reasoning in logic is the process of proving a new statement (conclusion) from other statements that are given and assumed to be true (the premises).^[80] Proofs can be structured as proof trees, in which nodes are labelled by sentences, and children nodes are connected to parent nodes by inference rules.

Given a problem and a set of premises, problem-solving reduces to searching for a proof tree whose root node is labelled by a solution of the problem and whose leaf nodes are labelled by premises or axioms. In the case of Horn clauses, problem-solving search can be performed by reasoning forwards from the premises or backwards from the problem.^[81] In the more general case of the clausal form of first-order logic, resolution is a single, axiom-free rule of inference, in which a problem is solved by proving a contradiction from premises that include the negation of the problem to be solved.^[82]

Inference in both Horn clause logic and first-order logic is undecidable, and therefore intractable. However, backward reasoning with Horn clauses, which underpins computation in the logic programming language Prolog, is Turing complete. Moreover, its efficiency is competitive with computation in other symbolic programming languages.^[83]

Fuzzy logic assigns a "degree of truth" between 0 and 1. It can therefore handle propositions that are vague and partially true.^[84]

Non-monotonic logics, including logic programming with negation as failure, are designed to handle default reasoning.^[26] Other specialized versions of logic have been developed to describe many complex domains.

Probabilistic methods for uncertain reasoning

Many problems in AI (including reasoning, planning, learning, perception, and robotics) require the agent to operate with incomplete or uncertain information. AI researchers have devised a number of tools to solve these problems using methods from probability theory and economics.^[85] Precise mathematical tools have been developed that analyze how an agent can make choices and plan, using decision theory, decision analysis,^[86] and information value theory.^[87] These tools include models such as Markov decision processes,^[88] dynamic decision networks,^[89] game theory and mechanism design.^[90]

Bayesian networks^[91] are a tool that can be used for reasoning (using the Bayesian inference algorithm),^{[g][93]} learning (using the expectation-maximization algorithm),^{[h][95]} planning (using decision networks)^[96] and perception (using dynamic Bayesian networks).^[89]

Probabilistic algorithms can also be used for filtering, prediction, smoothing, and finding explanations for streams of data, thus helping perception systems analyze processes that occur over time (e.g., hidden Markov models or Kalman filters).^[89]

Classifiers and statistical learning methods

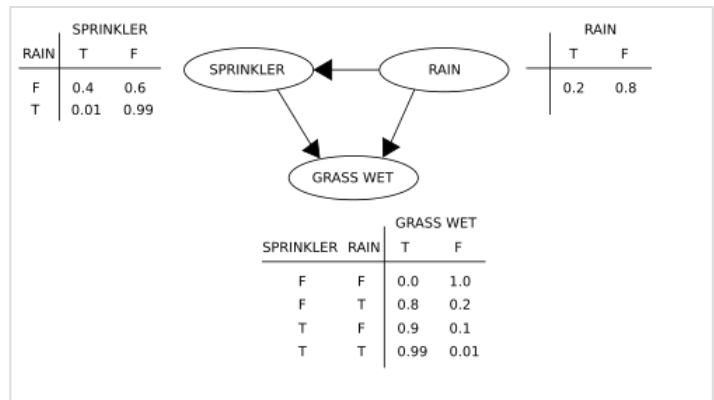
The simplest AI applications can be divided into two types: classifiers (e.g., "if shiny then diamond"), on one hand, and controllers (e.g., "if diamond then pick up"), on the other hand. Classifiers^[97] are functions that use pattern matching to determine the closest match. They can be fine-tuned based on chosen examples using supervised learning. Each pattern (also called an "observation") is labeled with a certain predefined class. All the observations combined with their class labels are known as a data set. When a new observation is received, that observation is classified based on previous experience.^[44]

There are many kinds of classifiers in use.^[98] The decision tree is the simplest and most widely used symbolic machine learning algorithm.^[99] K-nearest neighbor algorithm was the most widely used analogical AI until the mid-1990s, and Kernel methods such as the support vector machine (SVM) displaced k-nearest neighbor in the 1990s.^[100] The naive Bayes classifier is reportedly the "most widely used learner"^[101] at Google, due in part to its scalability.^[102] Neural networks are also used as classifiers.^[103]

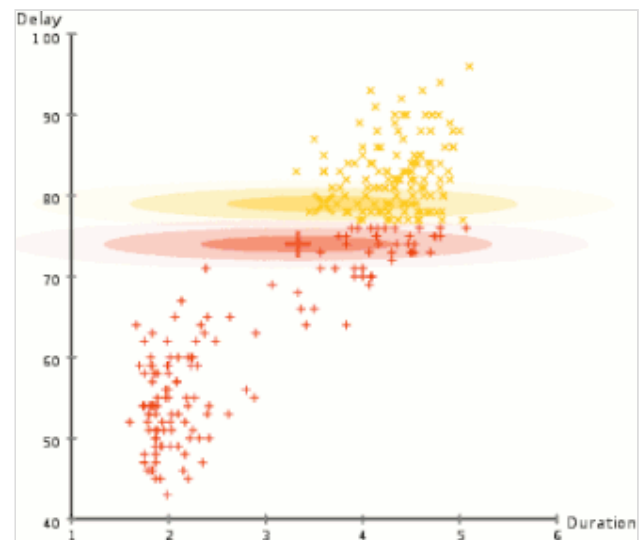
Artificial neural networks

An artificial neural network is based on a collection of nodes also known as artificial neurons, which loosely model the neurons in a biological brain. It is trained to recognise patterns; once trained, it can recognise those patterns in fresh data. There is an input, at least one hidden layer of nodes and an output. Each node applies a function and once the weight crosses its specified threshold, the data is transmitted to the next layer. A network is typically called a deep neural network if it has at least 2 hidden layers.^[103]

Learning algorithms for neural networks use local search to choose the weights that will get the right output for each input during training. The most common training technique is the backpropagation algorithm.^[104] Neural networks learn to model complex relationships between inputs and outputs and find patterns in data. In theory, a neural network can learn any function.^[105]

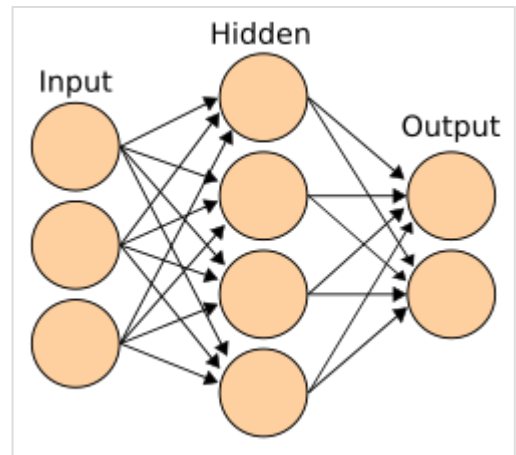


A simple Bayesian network, with the associated conditional probability tables



Expectation-maximization clustering of Old Faithful eruption data starts from a random guess but then successfully converges on an accurate clustering of the two physically distinct modes of eruption.

In feedforward neural networks the signal passes in only one direction.^[106] The term perceptron typically refers to a single-layer neural network.^[107] In contrast, deep learning uses many layers.^[108] Recurrent neural networks (RNNs) feed the output signal back into the input, which allows short-term memories of previous input events. Long short-term memory networks (LSTMs) are recurrent neural networks that better preserve longterm dependencies and are less sensitive to the vanishing gradient problem.^[109] Convolutional neural networks (CNNs) use layers of kernels to more efficiently process local patterns. This local processing is especially important in image processing, where the early CNN layers typically identify simple local patterns such as edges and curves, with subsequent layers detecting more complex patterns like textures, and eventually whole objects.^[110]

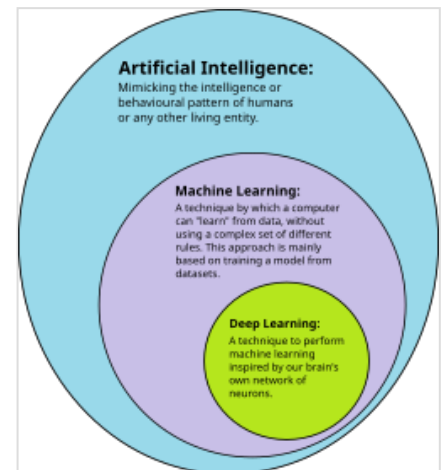


A neural network is an interconnected group of nodes, akin to the vast network of neurons in the human brain.

Deep learning

Deep learning uses several layers of neurons between the network's inputs and outputs.^[108] The multiple layers can progressively extract higher-level features from the raw input. For example, in image processing, lower layers may identify edges, while higher layers may identify the concepts relevant to a human such as digits, letters, or faces.^[112]

Deep learning has profoundly improved the performance of programs in many important subfields of artificial intelligence, including computer vision, speech recognition, natural language processing, image classification,^[113] and others. The reason that deep learning performs so well in so many applications is not known as of 2021.^[114] The sudden success of deep learning in 2012–2015 did not occur because of some new discovery or theoretical breakthrough (deep neural networks and backpropagation had been described by many people, as far back as the 1950s)^[i] but because of two factors: the incredible increase in computer power (including the hundred-fold increase in speed by switching to GPUs) and the availability of vast amounts of training data, especially the giant curated datasets used for benchmark testing, such as ImageNet.^[j]



Deep learning is a subset of machine learning, which is itself a subset of artificial intelligence.^[111]

GPT

Generative pre-trained transformers (GPT) are large language models (LLMs) that generate text based on the semantic relationships between words in sentences. Text-based GPT models are pre-trained on a large corpus of text that can be from the Internet. The pretraining consists of predicting the next token (a token being usually a word, subword, or punctuation). Throughout this pretraining, GPT models accumulate knowledge about the world and can then generate human-like text by repeatedly predicting the next token. Typically, a subsequent training phase makes the model more truthful, useful, and harmless, usually with a technique called reinforcement learning from human feedback (RLHF). Current GPT

models are prone to generating falsehoods called "hallucinations". These can be reduced with RLHF and quality data, but the problem has been getting worse for reasoning systems.^[122] Such systems are used in chatbots, which allow people to ask a question or request a task in simple text.^{[123][124]}

Current models and services include ChatGPT, Claude, Gemini, Copilot, and Meta AI.^[125] Multimodal GPT models can process different types of data (modalities) such as images, videos, sound, and text.^[126]

Hardware and software

In the late 2010s, graphics processing units (GPUs) that were increasingly designed with AI-specific enhancements and used with specialized TensorFlow software had replaced previously used central processing unit (CPUs) as the dominant means for large-scale (commercial and academic) machine learning models' training.^[127] Specialized programming languages such as Prolog were used in early AI research,^[128] but general-purpose programming languages like Python have become predominant.^[129]



Raspberry Pi AI Kit

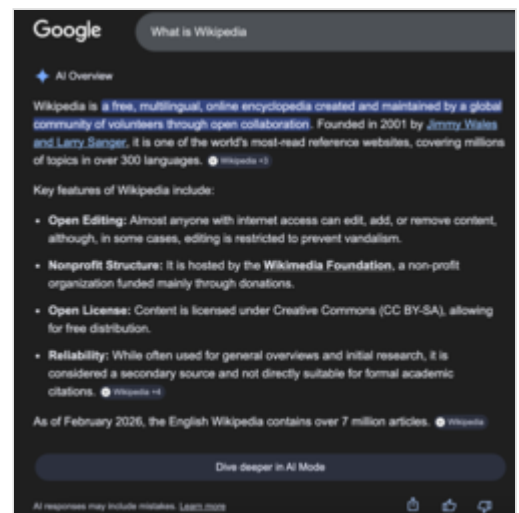
The transistor density in integrated circuits has been observed to roughly double every 18 months—a trend known as Moore's law, named after the Intel co-founder Gordon Moore, who first identified it. Improvements in GPUs have been even faster,^[130] a trend sometimes called Huang's law,^[131] named after Nvidia co-founder and CEO Jensen Huang.

Applications

AI and machine learning technology is used in most of the essential applications of the 2020s, including:

- search engines (such as Google Search)
- targeting online advertisements
- recommendation systems (offered by Netflix, YouTube or Amazon) driving internet traffic
- targeted advertising (AdSense, Facebook)
- virtual assistants (such as Siri or Alexa)
- autonomous vehicles (including drones, ADAS and self-driving cars)
- automatic language translation (Microsoft Translator, Google Translate)
- facial recognition (Apple's FaceID or Microsoft's DeepFace and Google's FaceNet)
- image labeling (used by Facebook, Apple's Photos and TikTok).

The deployment of AI may be overseen by a chief automation officer (CAO).



AI Overviews, an example of AI use on search engines

Health and medicine

It has been suggested that AI can overcome discrepancies in funding allocated to different fields of research.^[132]

AlphaFold 2 (2021) demonstrated the ability to approximate, in hours rather than months, the 3D structure of a protein.^[133] In 2023, it was reported that AI-guided drug discovery helped find a class of antibiotics capable of killing two different types of drug-resistant bacteria.^[134] In 2024, researchers used machine learning to accelerate the search for Parkinson's disease drug treatments. Their aim was to identify compounds that block the clumping, or aggregation, of alpha-synuclein (the protein that characterises Parkinson's disease). They were able to speed up the initial screening process ten-fold and reduce the cost by a thousand-fold.^{[135][136]}

Gaming

Game playing programs have been used since the 1950s to demonstrate and test AI's most advanced techniques.^[137] Deep Blue became the first computer chess-playing system to beat a reigning world chess champion, Garry Kasparov, on 11 May 1997.^[138] In 2011, in a Jeopardy! quiz show exhibition match, IBM's question answering system, Watson, defeated the two greatest Jeopardy! champions, Brad Rutter and Ken Jennings, by a significant margin.^[139] In March 2016, AlphaGo won 4 out of 5 games of Go in a match with Go champion Lee Sedol, becoming the first computer Go-playing system to beat a professional Go player without handicaps. Then, in 2017, it defeated Ke Jie, who was the best Go player in the world.^[140] Other programs handle imperfect-information games, such as the poker-playing program Pluribus.^[141] DeepMind developed increasingly generalistic reinforcement learning models, such as with MuZero, which could be trained to play chess, Go, or Atari games.^[142] In 2019, DeepMind's AlphaStar achieved grandmaster level in StarCraft II, a particularly challenging real-time strategy game that involves incomplete knowledge of what happens on the map.^[143] In 2021, an AI agent competed in a PlayStation Gran Turismo competition, winning against four of the world's best Gran Turismo drivers using deep reinforcement learning.^[144] In 2024, Google DeepMind introduced SIMA, a type of AI capable of autonomously playing nine previously unseen open-world video games by observing screen output, as well as executing short, specific tasks in response to natural language instructions.^[145]

Mathematics

In mathematics, probabilistic large language models are versatile, but can also produce wrong answers in the form of hallucinations. The Alibaba Group developed a version of its Qwen models called Qwen2-Math, that achieved state-of-the-art performance on several mathematical benchmarks, including 84% accuracy on the MATH dataset of competition mathematics problems.^[146] In January 2025, Microsoft proposed the technique rStar-Math that leverages Monte Carlo tree search and step-by-step reasoning, enabling a relatively small language model like Qwen-7B to solve 53% of the AIME 2024 and 90% of the MATH benchmark problems.^[147] Google DeepMind has developed models for solving mathematical problems: AlphaTensor, AlphaGeometry, AlphaProof and AlphaEvolve.^{[148][149]}

When natural language is used to describe mathematical problems, converters can transform such prompts into a formal language such as Lean to define mathematical tasks. The experimental model Gemini Deep Think accepts natural language prompts directly and achieved gold medal results in the International Math Olympiad of 2025.^[150]

Topological deep learning integrates various topological approaches.

Finance

According to Nicolas Firzli, director of the World Pensions & Investments Forum, it may be too early to see the emergence of highly innovative AI-informed financial products and services. He argues that "the deployment of AI tools will simply further automatise things: destroying tens of thousands of jobs in banking, financial planning, and pension advice in the process, but I'm not sure it will unleash a new wave of [e.g., sophisticated] pension innovation."^[151]

Military

Various countries are deploying AI military applications.^[152] The main applications enhance command and control, communications, sensors, integration and interoperability.^[153] Research is targeting intelligence collection and analysis, logistics, cyber operations, information operations, and semiautonomous and autonomous vehicles.^[152] AI technologies enable coordination of sensors and effectors, threat detection and identification, marking of enemy positions, target acquisition, coordination and deconfliction of distributed Joint Fires between networked combat vehicles, both human-operated and autonomous.^[153]

AI has been used in military operations in Iraq, Syria, Israel and Ukraine.^{[152][154][155][156]}

Generative AI

Generative artificial intelligence (GenAI) is a subfield of artificial intelligence (AI) that uses generative models to generate text, images, videos, audio, software code (vibe coding) or other forms of data.^[157] These models learn the underlying patterns and structures of their training data, and use them to generate new data^[158] in response to input, which often takes the form of natural language prompts.^{[159][160]}

The prevalence of generative AI tools has increased significantly since the AI boom in the 2020s. This boom was made possible by improvements in deep neural networks, particularly large language models (LLMs), which are based on the transformer architecture. Generative AI applications include chatbots such as ChatGPT, Claude, Copilot, DeepSeek, Google Gemini and Grok; text-to-image models such as DALL-E, Firefly, Stable Diffusion, and Midjourney; and text-to-video models such as Veo, LTX and Sora.^{[161][162][163]}

Companies in a variety of sectors have used generative AI, including those in software development, healthcare,^[164] finance,^[165] entertainment,^[166] customer service,^[167] sales and marketing,^[168] art, writing,^[169] and product design.^[170]

Agents

AI agents are software entities designed to perceive their environment, make decisions, and take actions autonomously to achieve specific goals. These agents can interact with users, their environment, or other agents. AI agents are used in various applications, including virtual assistants, chatbots, autonomous vehicles, game-playing systems, and industrial robotics. AI agents operate within the constraints of their programming, available computational resources, and hardware limitations. This means they are restricted to performing tasks within their defined scope and have finite memory and processing

capabilities. In real-world applications, AI agents often face time constraints for decision-making and action execution. Many AI agents incorporate learning algorithms, enabling them to improve their performance over time through experience or training. Using machine learning, AI agents can adapt to new situations and optimise their behaviour for their designated tasks.^{[171][172][173]}

Web search

Microsoft introduced Copilot Search in February 2023 under the name Bing Chat. Copilot Search provides AI-generated summaries.^[174]

Google introduced an AI Mode at its Google I/O event on 20 May 2025.^[175]

Sexuality

Applications of AI in this domain include AI-enabled menstruation and fertility trackers that analyze user data to offer predictions,^[176] AI-integrated sex toys (e.g., teledildonics),^[177] AI-generated sexual education content,^[178] and AI agents that simulate sexual and romantic partners (e.g., Replika).^[179] AI is also used for the production of non-consensual deepfake pornography, raising significant ethical and legal concerns.^[180]

AI technologies have also been used to attempt to identify online gender-based violence and online sexual grooming of minors.^{[181][182]}

Other industry-specific tasks

In a 2017 survey, one in five companies reported having incorporated "AI" in some offerings or processes.^[183]

In the field of evacuation and disaster management, AI has been used to investigate patterns in large-scale and small-scale evacuations using historical data from GPS, videos or social media.^{[184][185][186]}

During the 2024 Indian elections, US\$50 million was spent on authorized AI-generated content, notably by creating deepfakes of allied (including sometimes deceased) politicians to better engage with voters, and by translating speeches to various local languages.^[187]

The use of generative AI by law firms for legal research resulted in the creation of the global "AI Hallucination Cases" database, in April 2025, established by HEC Paris and Sciences Po legal data analysis lecturer Damien Charlotin.^{[188][189]} By 2026, judges had issued sanctions and bar associations had issued warnings due to attorney submissions to the courts containing fabricated case law citations hallucinated by AI tools.^[190]

Ethics

AI has potential benefits and potential risks.^[193] AI may be able to advance science and find solutions for serious problems: Demis Hassabis of DeepMind hopes to "solve intelligence, and then use that to solve everything else".^[194] However, as the use of AI has become widespread, several unintended

consequences and risks have been identified.^{[195][196]} In-production systems can sometimes not factor ethics and bias into their AI training processes, especially when the AI algorithms are inherently unexplainable in deep learning.^[197]

Risks and harm

Privacy and copyright

Machine learning algorithms require large amounts of data. The techniques used to acquire this data have raised concerns about privacy, surveillance and copyright.

AI-powered devices and services, such as virtual assistants and IoT products, continuously collect personal information, raising concerns about intrusive data gathering and unauthorized access by third parties. The loss of privacy is further exacerbated by AI's ability to process and combine vast amounts of data, potentially leading to a surveillance society where individual activities are constantly monitored and analyzed without adequate safeguards or transparency.

Sensitive user data collected may include online activity records, geolocation data, video, or audio.^[198] For example, in order to build speech recognition algorithms, Amazon has recorded millions of private conversations and allowed temporary workers to listen to and transcribe some of them.^[199] Opinions about this widespread surveillance range from those who see it as a necessary evil to those for whom it is clearly unethical and a violation of the right to privacy.^[200]

AI developers argue that this is the only way to deliver valuable applications and have developed several techniques that attempt to preserve privacy while still obtaining the data, such as data aggregation, de-identification and differential privacy.^[201] Since 2016, some privacy experts, such as Cynthia Dwork, have begun to view privacy in terms of fairness. Brian Christian wrote that experts have pivoted "from the question of 'what they know' to the question of 'what they're doing with it'".^[202]

Generative AI is often trained on unlicensed copyrighted works, including in domains such as images or computer code; the output is then used under the rationale of "fair use". Experts disagree about how well and under what circumstances this rationale will hold up in courts of law; relevant factors may include "the purpose and character of the use of the copyrighted work" and "the effect upon the potential market for the copyrighted work".^{[203][204]} Website owners can indicate that they do not want their content scraped via a "robots.txt" file.^[205] However, some companies will scrape content regardless^{[206][207]} because the robots.txt file has no real authority. In 2023, leading authors (including John Grisham and Jonathan Franzen) sued AI companies for using their work to train generative AI.^{[208][209]} Another discussed approach is to envision a separate sui generis system of protection for creations generated by AI to ensure fair attribution and compensation for human authors.^[210]



Street art in Tel Aviv^{[191][192]}

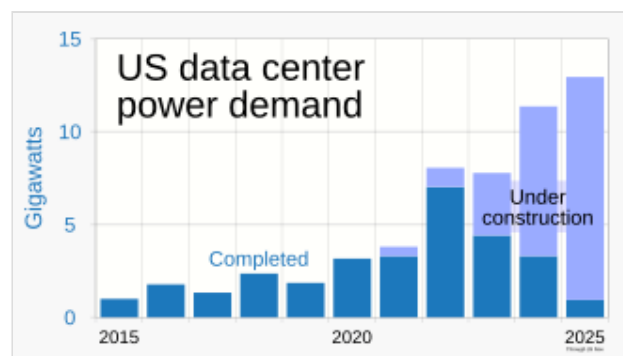
Dominance by tech giants

The commercial AI scene is dominated by Big Tech companies such as Alphabet Inc., Amazon, Apple Inc., Meta Platforms, and Microsoft.^{[211][212][213]} Some of these players already own the vast majority of existing cloud infrastructure and computing power from data centers, allowing them to entrench further in the marketplace.^{[214][215]}

Power needs and environmental impacts

Technology companies have built electricity and artificial intelligence infrastructure to facilitate the AI boom of the 2020s. A 2025 report from the consulting firm McKinsey & Company estimated that by 2030, \$2.7 trillion would be invested into AI infrastructure and data centers in the US, surpassing World War II's Manhattan Project every month.^[217]

In January 2024, the International Energy Agency (IEA) released *Electricity 2024, Analysis and Forecast to 2026*.^[218] This is the first IEA report to make projections for data centers and power consumption by AI and cryptocurrency. The report states that power demand for these uses might double by 2026, with the additional power consumption equaling that of Japan.^[219]



Fueled by a growth in AI, data centers' demand for power increased in the 2020s.^[216]

Power consumption by AI is responsible for an increase in fossil fuel use, and has delayed closings of obsolete, carbon-emitting coal energy facilities. A ChatGPT search involves the use of 10 times the electrical energy as a Google search.^[220]

A 2024 Goldman Sachs Research Paper, *AI Data Centers and the Coming US Power Demand Surge*, found "US power demand (is) likely to experience growth not seen in a generation...." and forecasts that, by 2030, US data centers will consume 8% of US power, as opposed to 3% in 2022, presaging growth for the electrical power generation industry by a variety of means.^[221] Data centers' need for more and more electrical power is such that they might max out the electrical grid. The Big Tech companies counter that AI can be used to maximize the utilization of the grid by all.^[222]

In 2024, *The Wall Street Journal* reported that big AI companies have begun negotiations with the US nuclear power providers to provide electricity to the data centers. In March 2024 Amazon purchased a Pennsylvania nuclear-powered data center for US\$650 million.^[223]

In September 2024, Microsoft announced an agreement with Constellation Energy to re-open the Three Mile Island nuclear power plant to provide Microsoft with 100% of all electric power produced by the plant for 20 years. Reopening the plant, which suffered a partial nuclear meltdown of its Unit 2 reactor in 1979, will require Constellation to get through strict regulatory processes which will include extensive safety scrutiny from the US Nuclear Regulatory Commission. If approved (this will be the first ever US re-commissioning of a nuclear plant), over 835 megawatts of power – enough for 800,000 homes – of energy will be produced. The cost for re-opening and upgrading is estimated at US\$1.6 billion and is dependent on tax breaks for nuclear power contained in the 2022 US Inflation Reduction Act.^[224] As of

2024, the US government and the state of Michigan have been investing almost US\$2 billion to reopen the Palisades Nuclear reactor on Lake Michigan. Closed since 2022, the plant was planned to be reopened in October 2025.^[225]

After the last approval in September 2023, Taiwan suspended the approval of data centers north of Taoyuan with a capacity of more than 5 MW in 2024, due to power supply shortages.^[226] Taiwan aims to phase out nuclear power by 2025.^[226]

Singapore imposed a ban on the opening of data centers in 2019 due to electric power, but in 2022, lifted this ban.^[226]

Although most nuclear plants in Japan have been shut down after the 2011 Fukushima nuclear accident, according to an October 2024 *Bloomberg* article in Japanese, cloud gaming services company Ubitus, in which Nvidia has a stake, is looking for land in Japan near a nuclear power plant for a new data center for generative AI.^[227]

On 1 November 2024, the Federal Energy Regulatory Commission (FERC) rejected an application submitted by Talen Energy for approval to supply some electricity from the nuclear power station Susquehanna to Amazon's data center.^[228] According to the Commission Chairman Willie L. Phillips, it is a burden on the electricity grid as well as a significant cost shifting concern to households and other business sectors.^[228]

In 2025, a report prepared by the IEA estimated the greenhouse gas emissions from the energy consumption of AI at 180 million tons. By 2035, these emissions could rise to 300–500 million tonnes depending on what measures will be taken. This is below 1.5% of the energy sector emissions. The emissions reduction potential of AI was estimated at 5% of the energy sector emissions, but rebound effects (for example if people switch from public transport to autonomous cars) can reduce it.^[229]

Misinformation

YouTube, Facebook and others use recommender systems to guide users to more content. These AI programs were given the goal of maximizing user engagement (that is, the only goal was to keep people watching). The AI learned that users tended to choose misinformation, conspiracy theories, and extreme partisan content, and, to keep them watching, the AI recommended more of it. Users also tended to watch more content on the same subject, so the AI led people into filter bubbles where they received multiple versions of the same misinformation.^[230] This convinced many users that the misinformation was true, and ultimately undermined trust in institutions, the media and the government.^[231] The AI program had correctly learned to maximize its goal, but the result was harmful to society. After the U.S. election in 2016, major technology companies took some steps to mitigate the problem.^[232]

In the early 2020s, generative AI began to create images, audio, and texts that are virtually indistinguishable from real photographs, recordings, or human writing,^[233] while realistic AI-generated videos became feasible in the mid-2020s.^{[234][235][236]} It is possible for bad actors to use this technology to create massive amounts of misinformation or propaganda;^[237] one such potential malicious use is deepfakes for computational propaganda.^[238] AI pioneer and Nobel Prize-winning computer scientist Geoffrey Hinton expressed concern about AI enabling "authoritarian leaders to manipulate their

electorates" on a large scale, among other risks.^[239] The ability to influence electorates has been proved in at least one study. This same study shows more inaccurate statements from the models when they advocate for candidates of the political right.^[240]

AI researchers at Microsoft, OpenAI, universities and other organisations have suggested using "personhood credentials" as a way to overcome online deception enabled by AI models.^[241]

Algorithmic bias and fairness

Machine learning applications can be biased^[k] if they learn from biased data.^[243] The developers may not be aware that the bias exists.^[244] Discriminatory behavior by some LLMs can be observed in their output.^[245] Bias can be introduced by the way training data is selected and by the way a model is deployed.^{[246][243]} If a biased algorithm is used to make decisions that can seriously harm people (as it can in medicine, finance, recruitment, housing or policing) then the algorithm may cause discrimination.^[247] The field of fairness studies how to prevent harms from algorithmic biases.

On 28 June 2015, Google Photos's new image labeling feature mistakenly identified Jacky Alcine and a friend as "gorillas" because they were black. The system was trained on a dataset that contained very few images of black people,^[248] a problem called "sample size disparity".^[249] Google "fixed" this problem by preventing the system from labelling *anything* as a "gorilla". Eight years later, in 2023, Google Photos still could not identify a gorilla, and neither could similar products from Apple, Facebook, Microsoft and Amazon.^[250]

COMPAS is a commercial program widely used by U.S. courts to assess the likelihood of a defendant becoming a recidivist. In 2016, Julia Angwin at ProPublica discovered that COMPAS exhibited racial bias, despite the fact that the program was not told the races of the defendants. Although the error rate for both whites and blacks was calibrated equal at exactly 61%, the errors for each race were different—the system consistently overestimated the chance that a black person would re-offend and would underestimate the chance that a white person would not re-offend.^[251] In 2017, several researchers^[l] showed that it was mathematically impossible for COMPAS to accommodate all possible measures of fairness when the base rates of re-offense were different for whites and blacks in the data.^[253]

A program can make biased decisions even if the data does not explicitly mention a problematic feature (such as "race" or "gender"). The feature will correlate with other features (like "address", "shopping history" or "first name"), and the program will make the same decisions based on these features as it would on "race" or "gender".^[254] Moritz Hardt said "the most robust fact in this research area is that fairness through blindness doesn't work."^[255]

Criticism of COMPAS highlighted that machine learning models are designed to make "predictions" that are only valid if we assume that the future will resemble the past. If they are trained on data that includes the results of racist decisions in the past, machine learning models must predict that racist decisions will be made in the future. If an application then uses these predictions as *recommendations*, some of these "recommendations" will likely be racist.^[256] Thus, machine learning is not well suited to help make decisions in areas where there is hope that the future will be *better* than the past. It is descriptive rather than prescriptive.^[m]

Bias and unfairness may go undetected because the developers are overwhelmingly white and male: among AI engineers, about 4% are black and 20% are women.^[249]

There are various conflicting definitions and mathematical models of fairness. These notions depend on ethical assumptions, and are influenced by beliefs about society. One broad category is distributive fairness, which focuses on the outcomes, often identifying groups and seeking to compensate for statistical disparities. Representational fairness tries to ensure that AI systems do not reinforce negative stereotypes or render certain groups invisible. Procedural fairness focuses on the decision process rather than the outcome. The most relevant notions of fairness may depend on the context, notably the type of AI application and the stakeholders. The subjectivity in the notions of bias and fairness makes it difficult for companies to operationalize them. Having access to sensitive attributes such as race or gender is also considered by many AI ethicists to be necessary in order to compensate for biases, but it may conflict with anti-discrimination laws.^[242]

At the 2022 ACM Conference on Fairness, Accountability, and Transparency a paper reported that a CLIP-based (Contrastive Language-Image Pre-training) robotic system reproduced harmful gender- and race-linked stereotypes in a simulated manipulation task. The authors recommended robot-learning methods which physically manifest such harms be "paused, reworked, or even wound down when appropriate, until outcomes can be proven safe, effective, and just."^{[258][259][260]}

Lack of transparency

Many AI systems are so complex that their designers cannot explain how they reach their decisions.^[261] Particularly with deep neural networks, in which there are many non-linear relationships between inputs and outputs. But some popular explainability techniques exist.^[262]

It is impossible to be certain that a program is operating correctly if no one knows how exactly it works. There have been many cases where a machine learning program passed rigorous tests, but nevertheless learned something different than what the programmers intended. For example, a system that could identify skin diseases better than medical professionals was found to actually have a strong tendency to classify images with a ruler as "cancerous", because pictures of malignancies typically include a ruler to show the scale.^[263] Another machine learning system designed to help effectively allocate medical resources was found to classify patients with asthma as being at "low risk" of dying from pneumonia. Having asthma is actually a severe risk factor, but since the patients having asthma would usually get much more medical care, they were relatively unlikely to die according to the training data. The correlation between asthma and low risk of dying from pneumonia was real, but misleading.^[264]

People who have been harmed by an algorithm's decision have a right to an explanation.^[265] Doctors, for example, are expected to clearly and completely explain to their colleagues the reasoning behind any decision they make. Early drafts of the European Union's General Data Protection Regulation in 2016 included an explicit statement that this right exists.^[n] Industry experts noted that this is an unsolved problem with no solution in sight. Regulators argued that nevertheless the harm is real: if the problem has no solution, the tools should not be used.^[266]

DARPA established the XAI ("Explainable Artificial Intelligence") program in 2014 to try to solve these problems.^[267]

Several approaches aim to address the transparency problem. SHAP enables to visualise the contribution of each feature to the output.^[268] LIME can locally approximate a model's outputs with a simpler, interpretable model.^[269] Multitask learning provides a large number of outputs in addition to the target classification. These other outputs can help developers deduce what the network has learned.^[270]

Deconvolution, DeepDream and other generative methods can allow developers to see what different layers of a deep network for computer vision have learned, and produce output that can suggest what the network is learning.^[271] For generative pre-trained transformers, Anthropic developed a technique based on dictionary learning that associates patterns of neuron activations with human-understandable concepts.^[272]

Bad actors and weaponized AI

Artificial intelligence provides a number of tools that are useful to bad actors, such as authoritarian governments, terrorists, criminals or rogue states.

A lethal autonomous weapon is a machine that locates, selects and engages human targets without human supervision.^[o] Widely available AI tools can be used by bad actors to develop inexpensive autonomous weapons and, if produced at scale, they are potentially weapons of mass destruction.^[274] Even when used in conventional warfare, they currently cannot reliably choose targets and could potentially kill an innocent person.^[274] In 2014, 30 nations (including China) supported a ban on autonomous weapons under the United Nations' Convention on Certain Conventional Weapons, however the United States and others disagreed.^[275] By 2015, over fifty countries were reported to be researching battlefield robots.^[276]

AI tools make it easier for authoritarian governments to efficiently control their citizens in several ways. Face and voice recognition allow widespread surveillance. Machine learning, operating this data, can classify potential enemies of the state and prevent them from hiding. Recommendation systems can precisely target propaganda and misinformation for maximum effect. Deepfakes and generative AI aid in producing misinformation. Advanced AI can make authoritarian centralized decision-making more competitive than liberal and decentralized systems such as markets. It lowers the cost and difficulty of digital warfare and advanced spyware.^[277] All these technologies have been available since 2020 or earlier—AI facial recognition systems are already being used for mass surveillance in China.^{[278][279]}

There are many other ways in which AI is expected to help bad actors, some of which can not be foreseen. For example, machine-learning AI is able to design tens of thousands of toxic molecules in a matter of hours.^[280]

Technological unemployment

Economists have frequently highlighted the risks of redundancies from AI, and speculated about unemployment if there is no adequate social policy for full employment.^[281]

In the past, technology has tended to increase rather than reduce total employment, but economists acknowledge that "we're in uncharted territory" with AI.^[282] A survey of economists showed disagreement about whether the increasing use of robots and AI will cause a substantial increase in long-term unemployment, but they generally agree that it could be a net benefit if productivity gains are redistributed.^[283] Risk estimates vary; for example, in the 2010s, Michael Osborne and Carl Benedikt Frey estimated 47% of U.S. jobs are at "high risk" of potential automation, while an OECD report classified only 9% of U.S. jobs as "high risk".^{[p][285]} The methodology of speculating about future employment levels has been criticised as lacking evidential foundation, and for implying that technology, rather than social policy, creates unemployment, as opposed to redundancies.^[281] In April 2023, it was

reported that 70% of the jobs for Chinese video game illustrators had been eliminated by generative artificial intelligence.^{[286][287]} Early-career workers showed decreasing employment rates in some AI-exposed occupations.^[288]

Unlike previous waves of automation, many middle-class jobs may be eliminated by artificial intelligence; *The Economist* stated in 2015 that "the worry that AI could do to white-collar jobs what steam power did to blue-collar ones during the Industrial Revolution" is "worth taking seriously".^[289] Jobs at extreme risk range from paralegals to fast food cooks, while job demand is likely to increase for care-related professions ranging from personal healthcare to the clergy.^[290] In July 2025, Ford CEO Jim Farley predicted that "artificial intelligence is going to replace literally half of all white-collar workers in the U.S."^[291]

From the early days of the development of artificial intelligence, there have been arguments, for example, those put forward by Joseph Weizenbaum, about whether tasks that can be done by computers actually should be done by them, given the difference between computers and humans, and between quantitative calculation and qualitative, value-based judgement.^[292]

Substitution for human–human interaction

With the increase of loneliness in the early 21st century, AI is sometimes identified as a potential source of relief to this problem. It would be possible, via human-like qualities built into AI products,^[293] for individuals to assume that this need can be met by artificial means.^{[294][295]} In some cases, people approach artificial intelligence for companionship when they believe that they would not find acceptance due to feeling outcast.^[296] Examples of harm coming to humans from advanced chatbots have been reported in courts in the United States, with AI companies accused of creating products that endanger humans through emotional confusion or deception.^{[297][298]}

Existential risk

Recent public debates in artificial intelligence have increasingly focused on its broader societal and ethical implications. It has been argued AI will become so powerful that humanity may irreversibly lose control of it. This could, as physicist Stephen Hawking stated, "spell the end of the human race".^[299] This scenario has been common in science fiction, when a computer or robot suddenly develops a human-like "self-awareness" (or "sentience" or "consciousness") and becomes a malevolent character.^[q] These sci-fi scenarios are misleading in several ways.

First, AI does not require human-like sentience to be an existential risk. Modern AI programs are given specific goals and use learning and intelligence to achieve them. Philosopher Nick Bostrom argued that if one gives *almost any* goal to a sufficiently powerful AI, it may choose to destroy humanity to achieve it (he used the example of an automated paperclip factory that destroys the world to get more iron for paperclips).^[301] Stuart Russell gives the example of household robot that tries to find a way to kill its owner to prevent it from being unplugged, reasoning that "you can't fetch the coffee if you're dead."^[302] In order to be safe for humanity, a superintelligence would have to be genuinely aligned with humanity's morality and values so that it is "fundamentally on our side".^[303]

Second, Yuval Noah Harari argues that AI does not require a robot body or physical control to pose an existential risk. The essential parts of civilization are not physical. Things like ideologies, law, government, money and the economy are built on language; they exist because there are stories that

billions of people believe. The current prevalence of misinformation suggests that an AI could use language to convince people to believe anything, even to take actions that are destructive.^[304] Geoffrey Hinton said in 2025 that modern AI is particularly "good at persuasion" and getting better all the time. He asks "Suppose you wanted to invade the capital of the US. Do you have to go there and do it yourself? No. You just have to be good at persuasion."^[305]

The opinions amongst experts and industry insiders are mixed, with sizable fractions both concerned and unconcerned by risk from eventual superintelligent AI.^[306] Personalities such as Stephen Hawking, Bill Gates, and Elon Musk,^[307] as well as AI pioneers such as Geoffrey Hinton, Yoshua Bengio, Stuart Russell, Demis Hassabis, and Sam Altman, have expressed concerns about existential risk from AI.

In May 2023, Geoffrey Hinton announced his resignation from Google in order to be able to "freely speak out about the risks of AI" without "considering how this impacts Google".^[308] He notably mentioned risks of an AI takeover,^[309] and stressed that in order to avoid the worst outcomes, establishing safety guidelines will require cooperation among those competing in use of AI.^[310]

In 2023, many leading AI experts endorsed the joint statement that "Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war".^[311]

Some other researchers were more optimistic. AI pioneer Jürgen Schmidhuber did not sign the joint statement, emphasising that in 95% of all cases, AI research is about making "human lives longer and healthier and easier."^[312] While the tools that are now being used to improve lives can also be used by bad actors, "they can also be used against the bad actors."^{[313][314]} Andrew Ng also argued that "it's a mistake to fall for the doomsday hype on AI—and that regulators who do will only benefit vested interests."^[315] Yann LeCun, a Turing Award winner, disagreed with the idea that AI will subordinate humans "simply because they are smarter, let alone destroy [us]",^[316] "scoff[ing] at his peers' dystopian scenarios of supercharged misinformation and even, eventually, human extinction." In contrast, he claimed that "intelligent machines will usher in a new renaissance for humanity, a new era of enlightenment."^[317] In the early 2010s, experts argued that the risks are too distant in the future to warrant research or that humans will be valuable from the perspective of a superintelligent machine.^[318] However, after 2016, the study of current and future risks and possible solutions became a serious area of research.^[319]

Ethical machines and alignment

Friendly AI are machines that have been designed from the beginning to minimize risks and to make choices that benefit humans. Eliezer Yudkowsky, who coined the term, argues that developing friendly AI should be a higher research priority: it may require a large investment and it must be completed before AI becomes an existential risk.^[320]

Machines with intelligence have the potential to use their intelligence to make ethical decisions. The field of machine ethics provides machines with ethical principles and procedures for resolving ethical dilemmas.^[321] The field of machine ethics is also called computational morality,^[321] and was founded at an AAAI symposium in 2005.^[322]

Other approaches include Wendell Wallach's "artificial moral agents"^[323] and Stuart J. Russell's three principles for developing provably beneficial machines.^[324]

Open source

Active organizations in the AI open-source community include Hugging Face,^[325] Google,^[326] EleutherAI and Meta.^[327] Various AI models, such as Llama 2, Mistral or Stable Diffusion, have been made open-weight,^{[328][329]} meaning that their architecture and trained parameters (the "weights") are publicly available. Open-weight models can be freely fine-tuned, which allows companies to specialize them with their own data and for their own use-case.^[330] Open-weight models are useful for research and innovation but can also be misused. Since they can be fine-tuned, any built-in security measure, such as objecting to harmful requests, can be trained away until it becomes ineffective. Some researchers warn that future AI models may develop dangerous capabilities (such as the potential to drastically facilitate bioterrorism) and that once released on the Internet, they cannot be deleted everywhere if needed. They recommend pre-release audits and cost-benefit analyses.^[331]

Frameworks

Artificial intelligence projects can be guided by ethical considerations during the design, development, and implementation of an AI system. An AI framework such as the Care and Act Framework, developed by the Alan Turing Institute and based on the SUM values, outlines four main ethical dimensions, defined as follows:^{[332][333]}

- **Respect** the dignity of individual people
- **Connect** with other people sincerely, openly, and inclusively
- **Care** for the wellbeing of everyone
- **Protect** social values, justice, and the public interest

Other developments in ethical frameworks include those decided upon during the Asilomar Conference, the Montreal Declaration for Responsible AI, and the IEEE's Ethics of Autonomous Systems initiative, among others;^[334] however, these principles are not without criticism, especially regarding the people chosen to contribute to these frameworks.^[335]

Promotion of the wellbeing of the people and communities that these technologies affect requires consideration of the social and ethical implications at all stages of AI system design, development and implementation, and collaboration between job roles such as data scientists, product managers, data engineers, domain experts, and delivery managers.^[336]

The UK AI Safety Institute released in 2024 a testing toolset called 'Inspect' for AI safety evaluations available under an MIT open-source licence which is freely available on GitHub and can be improved with third-party packages. It can be used to evaluate AI models in a range of areas including core knowledge, ability to reason, and autonomous capabilities.^[337]

Regulation

The regulation of artificial intelligence is the development of public sector policies and laws for promoting and regulating AI; it is therefore related to the broader regulation of algorithms.^[338] The regulatory and policy landscape for AI is an emerging issue in jurisdictions globally.^[339] According to AI Index at Stanford, the annual number of AI-related laws passed in the 127 survey countries jumped from one passed in 2016 to 37 passed in 2022 alone.^{[340][341]} Between 2016 and 2020, more than 30 countries adopted dedicated strategies for AI.^[342] Most EU member states had released national AI strategies, as

had Canada, China, India, Japan, Mauritius, the Russian Federation, Saudi Arabia, United Arab Emirates, U.S., and Vietnam. Others were in the process of elaborating their own AI strategy, including Bangladesh, Malaysia and Tunisia.^[342] The Global Partnership on Artificial Intelligence was launched in June 2020, stating a need for AI to be developed in accordance with human rights and democratic values, to ensure public confidence and trust in the technology.^[342] Henry Kissinger, Eric Schmidt, and Daniel Huttenlocher published a joint statement in November 2021 calling for a government commission to regulate AI.^[343] In 2023, OpenAI leaders published recommendations for the governance of superintelligence, which they believe may happen in less than 10 years.^[344] In 2023, the United Nations also launched an advisory body to provide recommendations on AI governance; the body comprises technology company executives, government officials and academics.^[345] On 1 August 2024, the EU Artificial Intelligence Act entered into force, establishing the first comprehensive EU-wide AI regulation.^[346] In 2024, the Council of Europe created the first international legally binding treaty on AI, called the "Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law". It was adopted by the European Union, the United States, the United Kingdom, and other signatories.^[347]



The first global AI Safety Summit was held in the United Kingdom in November 2023 with a declaration calling for international cooperation.

In a 2022 Ipsos survey, attitudes towards AI varied greatly by country; 78% of Chinese citizens, but only 35% of Americans, agreed that "products and services using AI have more benefits than drawbacks".^[340] A 2023 Reuters/Ipsos poll found that 61% of Americans agree, and 22% disagree, that AI poses risks to humanity.^[348] In a 2023 Fox News poll, 35% of Americans thought it "very important", and an additional 41% thought it "somewhat important", for the federal government to regulate AI, versus 13% responding "not very important" and 8% responding "not at all important".^{[349][350]}

In November 2023, the first global AI Safety Summit was held in Bletchley Park in the UK to discuss the near and far term risks of AI and the possibility of mandatory and voluntary regulatory frameworks.^[351] 28 countries including the United States, China, and the European Union issued a declaration at the start of the summit, calling for international co-operation to manage the challenges and risks of artificial intelligence.^{[352][353]} In May 2024 at the AI Seoul Summit, 16 global AI tech companies agreed to safety commitments on the development of AI.^{[354][355]}

In March 2026, the United Nations convened the inaugural meeting of the Independent International Scientific Panel on AI, a 40-member expert body established under the Global Digital Compact to produce annual evidence-based reports on AI's societal impacts.^[356]

History

The study of mechanical or "formal" reasoning began with philosophers and mathematicians in antiquity. The study of logic led directly to Alan Turing's theory of computation, which suggested that a machine, by shuffling symbols as simple as "0" and "1", could simulate any conceivable form of mathematical

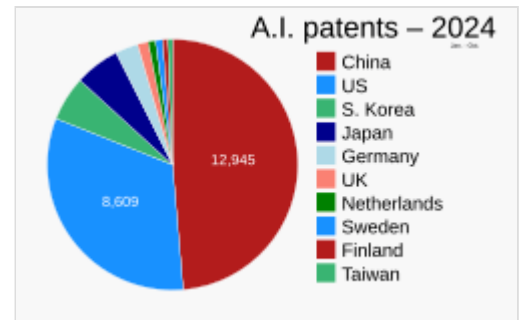
reasoning.^{[358][359]} This, along with concurrent discoveries in cybernetics, information theory and neurobiology, led researchers to consider the possibility of building an "electronic brain".^[r] They developed several areas of research that would become part of AI,^[361] such as McCulloch and Pitts design for "artificial neurons" in 1943,^[115] and Turing's influential 1950 paper 'Computing Machinery and Intelligence', which introduced the Turing test and showed that "machine intelligence" was plausible.^{[362][359]}

The field of AI research was founded at a workshop at Dartmouth College in 1956.^{[s][4]} The first AI program, Logic Theorist, was presented at the workshop, created by future Turing Award winner Allen Newell and future Nobel Laureate Herbert A. Simon, in collaboration with J. C. Shaw. Many of the workshop attendees became the leaders of AI research in the 1960s.^[t] They and their students produced programs that the press described as "astonishing":^[u] computers were learning checkers strategies, solving word problems in algebra, proving logical theorems and speaking English.^{[v][5]} Artificial intelligence laboratories were set up at a number of British and U.S. universities in the latter 1950s and early 1960s.^[359]

Researchers in the 1960s and the 1970s were convinced that their methods would eventually succeed in creating a machine with general intelligence and considered this the goal of their field.^[366] In 1965 Herbert Simon predicted, "machines will be capable, within twenty years, of doing any work a man can do".^[367] In 1967 Marvin Minsky agreed, writing that "within a generation ... the problem of creating 'artificial intelligence' will substantially be solved".^[368] They had, however, underestimated the difficulty of the problem.^[w] In 1974, both the U.S. and British governments cut off exploratory research in response to the criticism of Sir James Lighthill^[370] and ongoing pressure from the U.S. Congress to fund more productive projects.^[371] Minsky and Papert's book *Perceptrons* was understood as proving that artificial neural networks would never be useful for solving real-world tasks, thus discrediting the approach altogether.^[372] The "AI winter", a period when obtaining funding for AI projects was difficult, followed.^[7]

In the early 1980s, AI research was revived by the commercial success of expert systems,^[373] a form of AI program that simulated the knowledge and analytical skills of human experts. By 1985, the market for AI had reached over a billion dollars. At the same time, Japan's fifth generation computer project inspired the U.S. and British governments to restore funding for academic research.^[6] However, beginning with the collapse of the Lisp Machine market in 1987, AI once again fell into disrepute, and a second, longer-lasting winter began.^[8]

Up to this point, most of AI's funding had gone to projects that used high-level symbols to represent mental objects like plans, goals, beliefs, and known facts. In the 1980s, some researchers began to doubt that this approach would be able to imitate all the processes of human cognition, especially perception, robotics, learning and pattern recognition,^[374] and began to look into "sub-symbolic" approaches.^[375] Rodney Brooks rejected "representation" in general and focussed directly on engineering machines that move and survive.^[x] Judea Pearl, Lotfi Zadeh, and others developed methods that handled incomplete and uncertain information by making reasonable guesses rather than precise logic.^{[85][380]} But the most



In 2024, AI patents in China and the US numbered more than three-fourths of AI patents worldwide.^[357] Though China had more AI patents, the US had 35% more patents per AI patent-applicant company than China.^[357]

important development was the revival of "connectionism", including neural network research, by Geoffrey Hinton and others.^[381] In 1990, Yann LeCun successfully showed that convolutional neural networks can recognize handwritten digits, the first of many successful applications of neural networks.^[382]

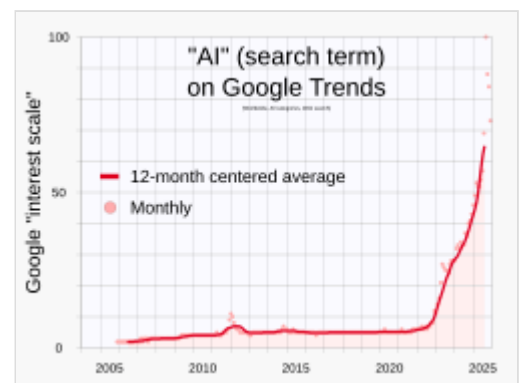
AI gradually restored its reputation in the late 1990s and early 21st century by exploiting formal mathematical methods and by finding specific solutions to specific problems. This "narrow" and "formal" focus allowed researchers to produce verifiable results and collaborate with other fields (such as statistics, economics and mathematics).^[383] By 2000, solutions developed by AI researchers were being widely used, although in the 1990s they were rarely described as "artificial intelligence" (a tendency known as the AI effect).^[384] However, several academic researchers became concerned that AI was no longer pursuing its original goal of creating versatile, fully intelligent machines. Beginning around 2002, they founded the subfield of artificial general intelligence (or "AGI"), which had several well-funded institutions by the 2010s.^[67]

Deep learning began to dominate industry benchmarks in 2012 and was adopted throughout the field.^[9] For many specific tasks, other methods were abandoned.^[y] Deep learning's success was based on both hardware improvements (faster computers,^[386] graphics processing units, cloud computing^[387]) and access to large amounts of data^[388] (including curated datasets,^[387] such as ImageNet). Deep learning's success led to an enormous increase in interest and funding in AI.^[z] The amount of machine learning research (measured by total publications) increased by 50% in the years 2015–2019.^[342]

In 2016, issues of fairness and the misuse of technology were catapulted into center stage at machine learning conferences, publications vastly increased, funding became available, and many researchers re-focussed their careers on these issues. The alignment problem became a serious field of academic study.^[319]

In the late 2010s and early 2020s, AGI companies began to deliver programs that created enormous interest. In 2015, AlphaGo, developed by DeepMind, beat the world champion Go player. The program taught only the game's rules and developed a strategy by itself. GPT-3 is a large language model that was released in 2020 by OpenAI and is capable of

generating high-quality human-like text.^[389] ChatGPT, launched on 30 November 2022, became the fastest-growing consumer software application in history, gaining over 100 million users in two months.^[390] It marked what is widely regarded as AI's breakout year, bringing it into the public consciousness.^[391] These programs, and others, inspired an aggressive AI boom, where large companies began investing billions of dollars in AI research. According to AI Impacts, about US\$50 billion annually was invested in "AI" around 2022 in the U.S. alone and about 20% of the new U.S. Computer Science PhD graduates have specialized in "AI".^[392] About 800,000 "AI"-related U.S. job openings existed in 2022.^[393] According to PitchBook research, 22% of newly funded startups in 2024 claimed to be AI companies.^[394]



The number of Google searches for the term "AI" accelerated in 2022.

Philosophy

Philosophical debates have historically sought to determine the nature of intelligence and how to make intelligent machines.^[395] Another major focus has been whether machines can be conscious, and the associated ethical implications.^[396] Many other topics in philosophy are relevant to AI, such as epistemology and free will.^[397] Rapid advancements have intensified public discussions on the philosophy and ethics of AI.^[396]

Defining artificial intelligence

Alan Turing investigated whether machines can show intelligent behaviour and think. In 1950, he proposed the Turing test, which measures the ability of a machine to simulate human conversation.^{[398][362]} Since we can only observe the behavior of the machine, it does not matter if it is "actually" thinking or literally has a "mind". Turing notes that we can not determine these things about other people but "it is usual to have a polite convention that everyone thinks."^[399]

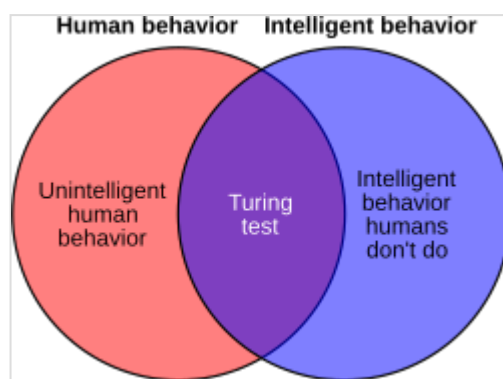
Russell and Norvig agree with Turing that intelligence must be defined in terms of external behavior, not internal structure.^[1] However, they are critical that the test requires the machine to imitate humans. "Aeronautical engineering texts", they wrote, "do not define the goal of their field as making 'machines that fly so exactly like pigeons that they can fool other pigeons.'"^[401] AI founder John McCarthy agreed, writing that "Artificial intelligence is not, by definition, simulation of human intelligence".^[402]

McCarthy defines intelligence as "the computational part of the ability to achieve goals in the world".^[403] Another AI founder, Marvin Minsky, similarly describes it as "the ability to solve hard problems".^[404] Artificial Intelligence: A Modern Approach defines it as the study of agents that perceive their environment and take actions that maximize their chances of achieving defined goals.^[1]

The many differing definitions of AI have been critically analyzed.^{[405][406][407]} During the 2020s AI boom, the term has been used as a marketing buzzword to promote products and services which do not use AI.^[408]

Legal definitions

The International Organization for Standardization describes an AI system as a "an engineered system that generates outputs such as content, forecasts, recommendations, or decisions for a given set of human-defined objectives, and can operate with varying levels of automation".^[409] The EU AI Act defines an AI system as "a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments".^[410] In the United



The Turing test can provide some evidence of intelligence, but it penalizes non-human intelligent behavior.^[400]

States, influential but non-binding guidance such as the National Institute of Standards and Technology's AI Risk Management Framework describes an AI system as "an engineered or machine-based system that can, for a given set of objectives, generate outputs such as predictions, recommendations, or decisions influencing real or virtual environments. AI systems are designed to operate with varying levels of autonomy".^[411]

Evaluating approaches to AI

No established unifying theory or paradigm has guided AI research for most of its history.^[aa] The unprecedented success of statistical machine learning in the 2010s eclipsed all other approaches (so much so that some sources, especially in the business world, use the term "artificial intelligence" to mean "machine learning with neural networks"). This approach is mostly sub-symbolic, soft and narrow. Critics argue that these questions may have to be revisited by future generations of AI researchers.

Symbolic AI and its limits

Symbolic AI (or "GOFAI")^[413] simulated the high-level conscious reasoning that people use when they solve puzzles, express legal reasoning and do mathematics. They were highly successful at "intelligent" tasks such as algebra or IQ tests. In the 1960s, Newell and Simon proposed the physical symbol systems hypothesis: "A physical symbol system has the necessary and sufficient means of general intelligent action."^[414]

However, the symbolic approach failed on many tasks that humans solve easily, such as learning, recognizing an object or commonsense reasoning. Moravec's paradox is the discovery that high-level "intelligent" tasks were easy for AI, but low level "instinctive" tasks were extremely difficult.^[415] Philosopher Hubert Dreyfus had argued since the 1960s that human expertise depends on unconscious instinct rather than conscious symbol manipulation, and on having a "feel" for the situation, rather than explicit symbolic knowledge.^[416] Although his arguments had been ridiculed and ignored when they were first presented, eventually, AI research came to agree with him.^{[ab][14]}

The issue is not resolved: sub-symbolic reasoning can make many of the same inscrutable mistakes that human intuition does, such as algorithmic bias. Critics such as Noam Chomsky argue continuing research into symbolic AI will still be necessary to attain general intelligence,^{[418][419]} in part because sub-symbolic AI is a move away from explainable AI: it can be difficult or impossible to understand why a modern statistical AI program made a particular decision. The emerging field of neuro-symbolic artificial intelligence attempts to bridge the two approaches.

Neat vs. scruffy

"Neats" hope that intelligent behavior is described using simple, elegant principles (such as logic, optimization, or neural networks). "Scruffies" expect that it necessarily requires solving a large number of unrelated problems. Neats defend their programs with theoretical rigor, scruffies rely mainly on incremental testing to see if they work. This issue was actively discussed in the 1970s and 1980s,^[420] but eventually was seen as irrelevant. Modern AI has elements of both.

Soft vs. hard computing

Finding a provably correct or optimal solution is intractable for many important problems.^[13] Soft computing is a set of techniques, including genetic algorithms, fuzzy logic and neural networks, that are tolerant of imprecision, uncertainty, partial truth and approximation. Soft computing was introduced in the late 1980s and most successful AI programs in the 21st century are examples of soft computing with neural networks.

Narrow vs. general AI

AI researchers are divided as to whether to pursue the goals of artificial general intelligence and superintelligence directly or to solve as many specific problems as possible (narrow AI) in hopes these solutions will lead indirectly to the field's long-term goals.^{[421][422]} General intelligence is difficult to define and difficult to measure, and modern AI has had more verifiable successes by focusing on specific problems with specific solutions. The sub-field of artificial general intelligence studies this area exclusively.

Machine consciousness, sentience, and mind

There is no settled consensus in philosophy of mind on whether a machine can have a mind, consciousness and mental states in the same sense that human beings do. This issue considers the internal experiences of the machine, rather than its external behavior. Mainstream AI research considers this issue irrelevant because it does not affect the goals of the field: to build machines that can solve problems using intelligence. Russell and Norvig add that "[t]he additional project of making a machine conscious in exactly the way humans are is not one that we are equipped to take on."^[423] However, the question has become central to the philosophy of mind. It is also typically the central question at issue in artificial intelligence in fiction.

Consciousness

David Chalmers identified two problems in understanding the mind, which he named the "hard" and "easy" problems of consciousness.^[424] The easy problem is understanding how the brain processes signals, makes plans and controls behavior. The hard problem is explaining how this *feels* or why it should feel like anything at all, assuming we are right in thinking that it truly does feel like something (Dennett's consciousness illusionism says this is an illusion). While human information processing is easy to explain, human subjective experience is difficult to explain. For example, it is easy to imagine a color-blind person who has learned to identify which objects in their field of view are red, but it is not clear what would be required for the person to *know what red looks like*.^[425]

Computationalism and functionalism

Computationalism is the position in the philosophy of mind that the human mind is an information processing system and that thinking is a form of computing. Computationalism argues that the relationship between mind and body is similar or identical to the relationship between software and hardware and thus may be a solution to the mind–body problem. This philosophical position was inspired by the work of AI researchers and cognitive scientists in the 1960s and was originally proposed by philosophers Jerry Fodor and Hilary Putnam.^[426]

Philosopher John Searle characterized this position as "strong AI": "The appropriately programmed computer with the right inputs and outputs would thereby have a mind in exactly the same sense human beings have minds."^[ac] Searle challenges this claim with his Chinese room argument, which attempts to show that even a computer capable of perfectly simulating human behavior would not have a mind.^[430]

AI welfare and rights

It is difficult or impossible to reliably evaluate whether an advanced AI is sentient (has the ability to feel), and if so, to what degree.^[431] But if there is a significant chance that a given machine can feel and suffer, then it may be entitled to certain rights or welfare protection measures, similarly to animals.^{[432][433]} Sapience (a set of capacities related to high intelligence, such as discernment or self-awareness) may provide another moral basis for AI rights.^[432] Robot rights are also sometimes proposed as a practical way to integrate autonomous agents into society.^[434]

In 2017, the European Union considered granting "electronic personhood" to some of the most capable AI systems. Similarly to the legal status of companies, it would have conferred rights but also responsibilities.^[435] Critics argued in 2018 that granting rights to AI systems would downplay the importance of human rights, and that legislation should focus on user needs rather than speculative futuristic scenarios. They also noted that robots lacked the autonomy to take part in society on their own.^{[436][437]}

Progress in AI increased interest in the topic. Proponents of AI welfare and rights often argue that AI sentience, if it emerges, would be particularly easy to deny. They warn that this may be a moral blind spot analogous to slavery or factory farming, which could lead to large-scale suffering if sentient AI is created and carelessly exploited.^{[433][432]}

Future

Superintelligence and the singularity

A superintelligence is a hypothetical agent that would possess intelligence far surpassing that of the brightest and most gifted human mind.^[422] If research into artificial general intelligence produced sufficiently intelligent software, it might be able to reprogram and improve itself. The improved software would be even better at improving itself, leading to what I. J. Good called an "intelligence explosion" and Vernor Vinge called a "singularity".^[438]

However, technologies cannot improve exponentially indefinitely, and typically follow an S-shaped curve, slowing when they reach the physical limits of what the technology can do.^[439]

Transhumanism

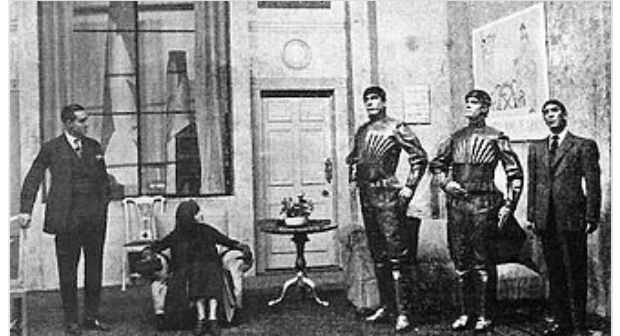
Robot designer Hans Moravec, cyberneticist Kevin Warwick and inventor Ray Kurzweil have predicted that humans and machines may merge in the future into cyborgs that are more capable and powerful than either. This idea, called transhumanism, has roots in the writings of Aldous Huxley and Robert Ettinger.^[440]

Edward Fredkin argues that "artificial intelligence is the next step in evolution", an idea first proposed by Samuel Butler's "Darwin among the Machines" as far back as 1863, and expanded upon by George Dyson in his 1998 book *Darwin Among the Machines: The Evolution of Global Intelligence*.^[441]

In fiction

Thought-capable artificial beings have appeared as storytelling devices since antiquity,^[442] and have been a persistent theme in science fiction.^[443]

A common trope in these works began with Mary Shelley's *Frankenstein*, where a human creation becomes a threat to its masters. This includes such works as Arthur C. Clarke's and Stanley Kubrick's *2001: A Space Odyssey* (both 1968), with HAL 9000, the murderous computer in charge of the *Discovery One* spaceship, as well as *Blade Runner* (1982), *The Terminator* (1984) and *The Matrix* (1999). In contrast, the rare loyal robots such as Gort from *The Day the Earth Stood Still* (1951) and Bishop from *Aliens* (1986) are less prominent in popular culture.^[444]



The word "robot" itself was coined by Karel Čapek in his 1921 play *R.U.R.*, the title standing for "Rossum's Universal Robots".

Isaac Asimov introduced the Three Laws of Robotics in many stories, most notably with the "Multivac" super-intelligent computer. Asimov's laws are often brought up during lay discussions of machine ethics,^[445] while almost all artificial intelligence researchers are familiar with Asimov's laws through popular culture, they generally consider the laws useless for many reasons, one of which is their ambiguity.^[446]

Several works use AI to force us to confront the fundamental question of what makes us human, showing us artificial beings that have the ability to feel, and thus to suffer. This appears in Karel Čapek's *R.U.R.*, the films *A.I. Artificial Intelligence* and *Ex Machina*, as well as the novel *Do Androids Dream of Electric Sheep?*, by Philip K. Dick. Dick considers the idea that our understanding of human subjectivity is altered by technology created with artificial intelligence.^[447]

See also

- Artificial consciousness – Hypothetical consciousness in artificial systems
- Artificial intelligence and elections
- Artificial intelligence content detection – Software to detect AI-generated content
- Artificial intelligence in Wikimedia projects – Use of artificial intelligence to develop Wikipedia and other Wikimedia projects
- Association for the Advancement of Artificial Intelligence (AAAI)
- Behavior selection algorithm – Algorithm that selects actions for intelligent agents
- Business process automation – Automation of business processes
- Case-based reasoning – Process of solving new problems based on the solutions of similar past problems